

Biomaterials

生物材料 (MSE338)

南方科技大学材料科学与工程系2024年春季课程

三教301, 周三9-10节

Instructor :

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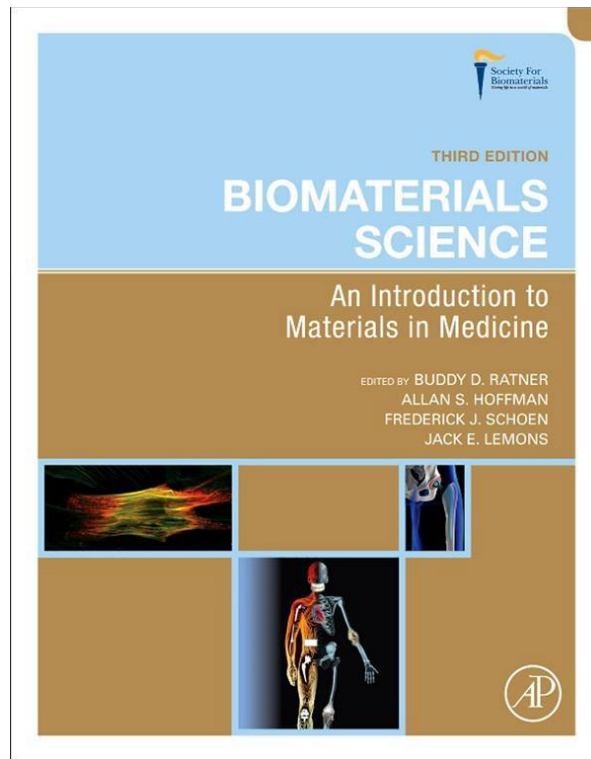
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Homework : 30%; Quiz : 20%; Project: 20%; Final Exam: 30%.

Text book:

Biomaterials Science, 3rd Edition, ISBN 9780123746269 , 2012.

References: Notes will be handed out during class.



Content	Topics	Notes
1	Properties of materials	
2	Introduction of various types of biomedical materials	
3	Surface modification of biomaterials	
4	Introduction of background knowledge and concepts in biology and medicine	
5	Host actions of biomaterials and their evaluations	
6	Degradation of biomaterials	
7	Applications of materials in medicine	
8	Final examination	

Properties of materials

A: General introduction

What is a biomaterial?

Applications and history of biomaterials.

Scientific and regulatory definition of biomaterials

B: Bulk and surface properties of materials

Bonds in materials

Characterization methods

What are biomaterials?

Any material of natural or of synthetic origin that comes in contact with tissue, blood or biological fluids, and intended for use in prosthetic, diagnostic, therapeutic or storage application

Course goals:

This course is an elective major course. It mainly introduces the principle, preparation, application and property of various common biomaterials in industry, as well as latest development of biomaterials research.

- ⌘ Differentiate various classes of biomaterials;
- ⌘ Differentiate various analytical methods used to characterize biomaterials;
- ⌘ Differentiate molecular and cellular events that follow exposure of materials to tissues and fluids;
- ⌘ Differentiate various biomedical devices; &
- ⌘ Describe various aspects of biomedical device design, fabrication and testing.

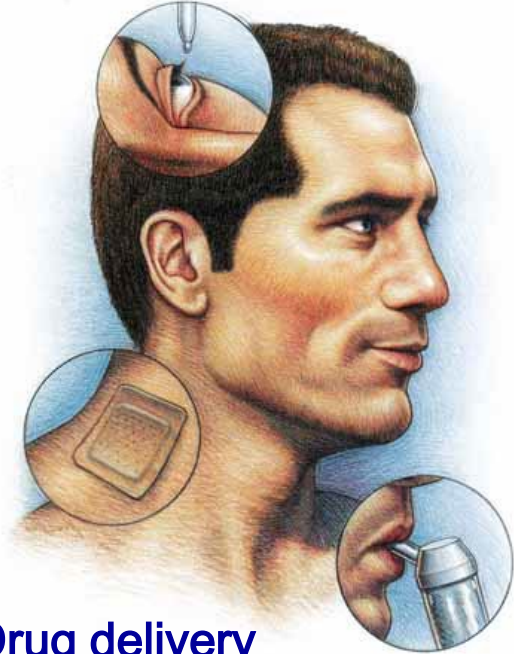
Biomaterials In Our Life



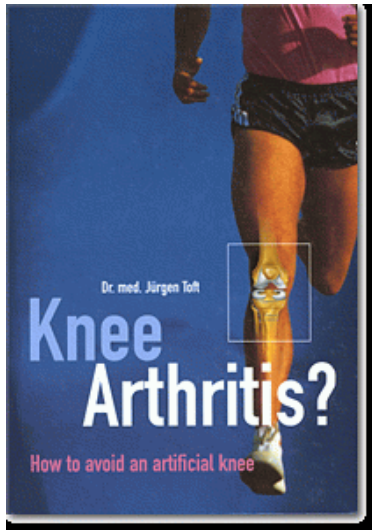
Lens



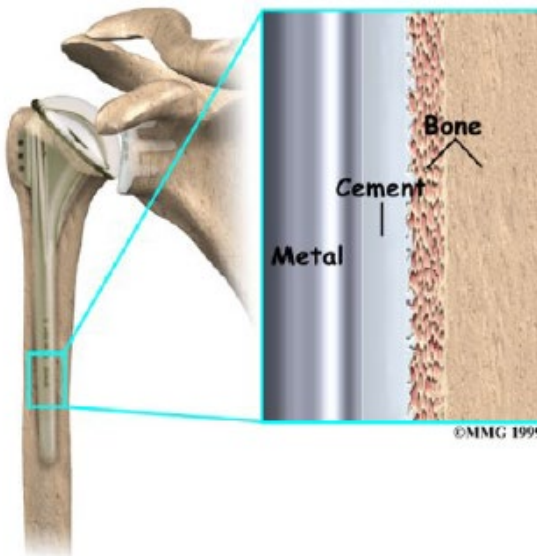
Heart



Drug delivery



Knee joint



Dentistry

Common Applications for Materials

Synthetic and Naturally Derived Polymers Commonly Used in Biomedical Applications

Polymer	Applications
Synthetic	
Poly(2-hydroxyethyl methacrylate)	Contact lenses
Poly(dimethyl siloxane)	Breast implants, contact lenses, knuckle replacements
Poly(ethylene)	Orthopedic joint implants
Poly(ethylene glycol)	Pharmaceutical fillers, wound dressings
Poly(ethylene terephthalate)	Vascular grafts, sutures
Poly(ϵ -caprolactone)	Drug delivery devices, sutures
Poly(lactic-co-glycolic acid)	Resorbable meshes and sutures
Poly(methyl methacrylate)	Bone cements, diagnostic contact lenses
Poly(tetrafluoroethylene)	Vascular grafts, sutures
Poly(isoprene)	Gloves
Poly(propylene)	Sutures
Naturally derived	
Alginate	Wound dressings
Chitosan	Wound dressings
Collagen	Orthopedic repair matrices, nerve repair matrices, tissue engineering matrices
Elastin	Skin repair matrices
Fibrin	Hemostatic products, tissue sealants
Glycosaminoglycan	Orthopedic repair matrices
Hyaluronic acid	Orthopedic repair matrices

Polymers

Suture: 缝合

Ceramics Commonly Used in Biomedical Applications

Ceramic	Applications
Aluminum oxides	Orthopedic joint replacement components, orthopedic load-bearing implants, implant coatings, dental implants
Bioactive glasses	Orthopedic and dental implant coatings, dental implants, facial reconstruction components, bone graft substitute materials
Calcium phosphates	Orthopedic and dental implant coatings, dental implant materials, bone graft substitute materials, bone cements

Ceramics

Metals

Metals Commonly Used in Biomedical Applications

Metal	Applications
Cobalt-chromium alloys	Artificial heart valves, dental prostheses, orthopedic fixation plates, artificial joint components, vascular stents
Gold and platinum	Dental fillings, electrodes for cochlear implants
Silver-tin-copper alloys	Dental amalgams
Stainless steel	Dental prostheses, orthopedic fixation plates, vascular stents
Titanium alloys	Artificial heart valves, dental implants, artificial joint components, orthopedic screws, pacemaker cases, vascular stents

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History of Biomaterials

~ 3000 B.C. : earliest report of a surgical suture in ancient Egypt.

~ 900 A.D. : estimated year of the first dental implant found in Europe, which was found to have properly integrated bone.

1829 : H.S. Levert studies canine responses to implanted metals “*in vivo*” – in the living body of a plant or animal.

•**1860**: British surgeon Joseph Lister introduced aseptic surgical techniques “aseptic” – free from contamination by harmful bacteria, viruses or other microorganisms



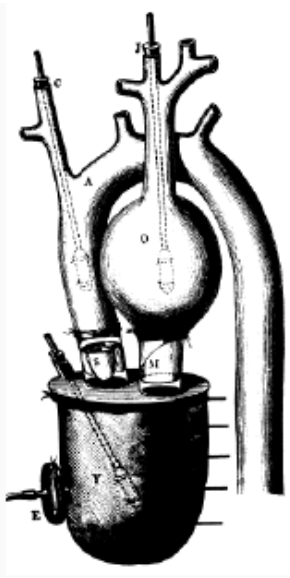
Canine metal implants, 1829



Lister in theatre

History of Biomaterials

- 1881** : French scientist and thinker, Étienne-Jule Marey, described an artificial heart device.
- 1886** : German doctor H. Hansmann is the first surgeon to use metal plates for internal fixation.
- 1931** : Boston surgeon Smith Peterson developed a metal cup for partial hip implants.



A design of artificial heart, 1881



Smith-Peterson cup

History of Biomaterials

- 1939 – 1945:** WWII spurs the development of many new materials and orthopaedic surgical techniques. Up until ~1950, mostly metals were used because very few plastics existed.
- 1947:** first paper on polyethylene as a synthetic implant material.
- 1950 - present:** a Biomaterials Revolution. New polymers, metals, ceramics have been produced, and biocompatibility on a *cellular* and *molecular* level were studied with increasingly better laboratory technologies.



Hip prostheses, 1956



Silicone contact lens, 2002

A look forward to recent technologies

Biomaterials research in the *modern* era is marked by a high degree of **interdisciplinarity**: ---

Materials science

--- **Chemistry and chemical engineering**

--- **Biology and bioengineering**

--- **Physics and biophysics**

--- **Biomechanics and mechanical engineering**

--- **Nanotechnology**

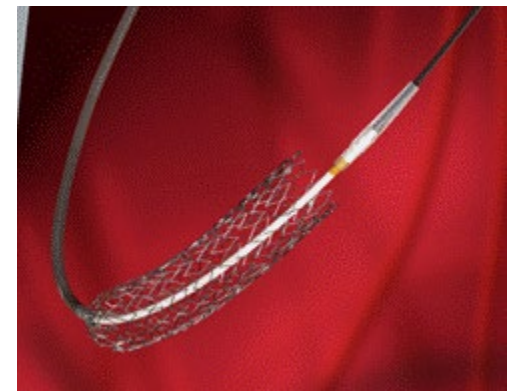


Example 1 – vascular stents

Boston Scientific's "Epic" vascular stent system

Granted marketing approval by FDA in 2012 Made from nitinol (nickel titanium alloy)

<http://www.youtube.com/watch?v=llGxRd4yLE8>



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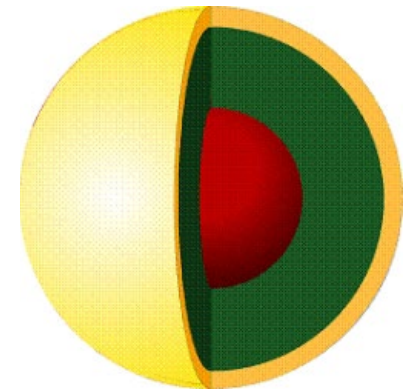
---Chemistry and chemical engineering

---Biology and bioengineering

---Physics and biophysics

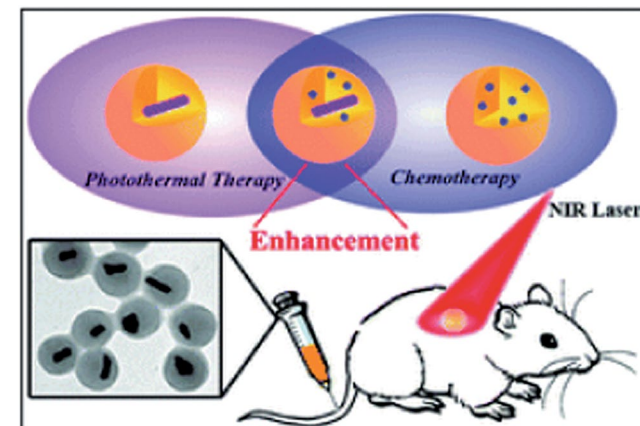
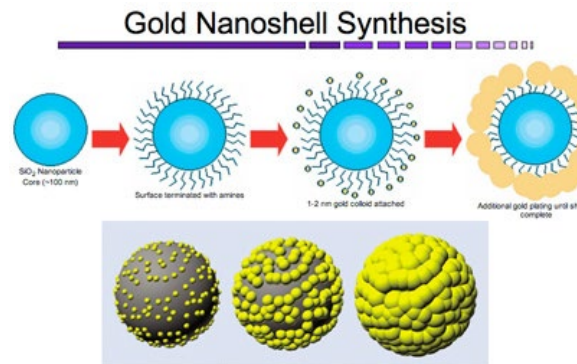
---Biomechanics and mechanical engine

---Nanotechnology



Example 2 – gold nanoshells

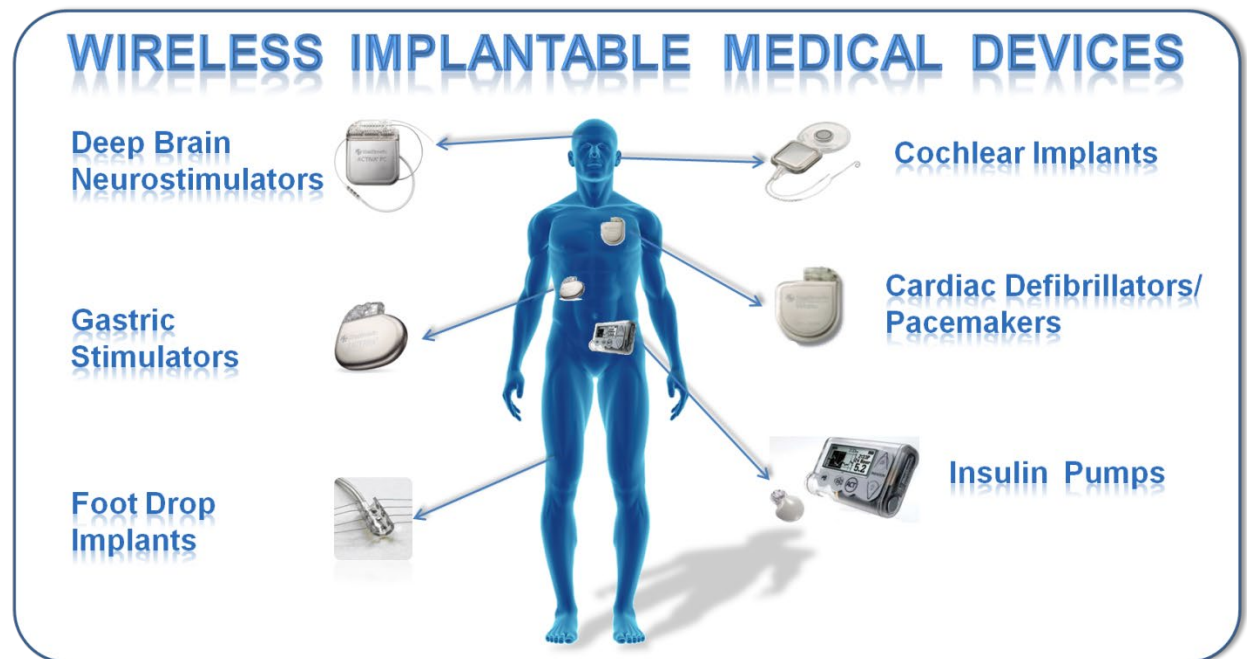
<https://presentations.acs.org/common/media-player.aspx/Spring2014/COLL/COLL07a/13154>



These two examples are given to illustrate how far we have come!

Biomaterials Research in Industry

is dominated as much by the regulatory approval process and submission requirements as by the physical, mechanical, and chemical properties of the medical device.



The Food and Drug Administration (www.fda.gov)

Regulates:

- Food
- Drugs-*Prescription, Over-the-Counter, Generic....*
- Medical Devices
- Animal Feed and Drugs-*Livestock, Pets...*
- Cosmetics-*Safety, Labeling.....*
- Radiation Emitting Products-*Cell Phones, Lasers, Microwaves.....*

Center for Devices and Radiologic Health- (CDRH)

- Responsible for regulating firms who manufacture, repackage, relabel, and/or import medical devices sold in the United States.
- Webpage: www.fda.gov/cdrh/



At the Center for Devices and Radiological Health (CDRH), clinical trials are the foundation for our decisions to approve the most important medical devices—products that have the potential to save or sustain life, but that also present the greatest risk to patients.

BIOMATERIAL OR MEDICAL DEVICE?

- In a very few instances, such biomaterials as injectable collagen, certain dental compounds, or bone cements are used as final products, **FDA regulates these materials as devices**
- the properties and safety of materials must be carefully assessed with respect to the specific application in question and its degree of patient contact.
- the final assessment must be performed on the finished product, under actual use conditions.

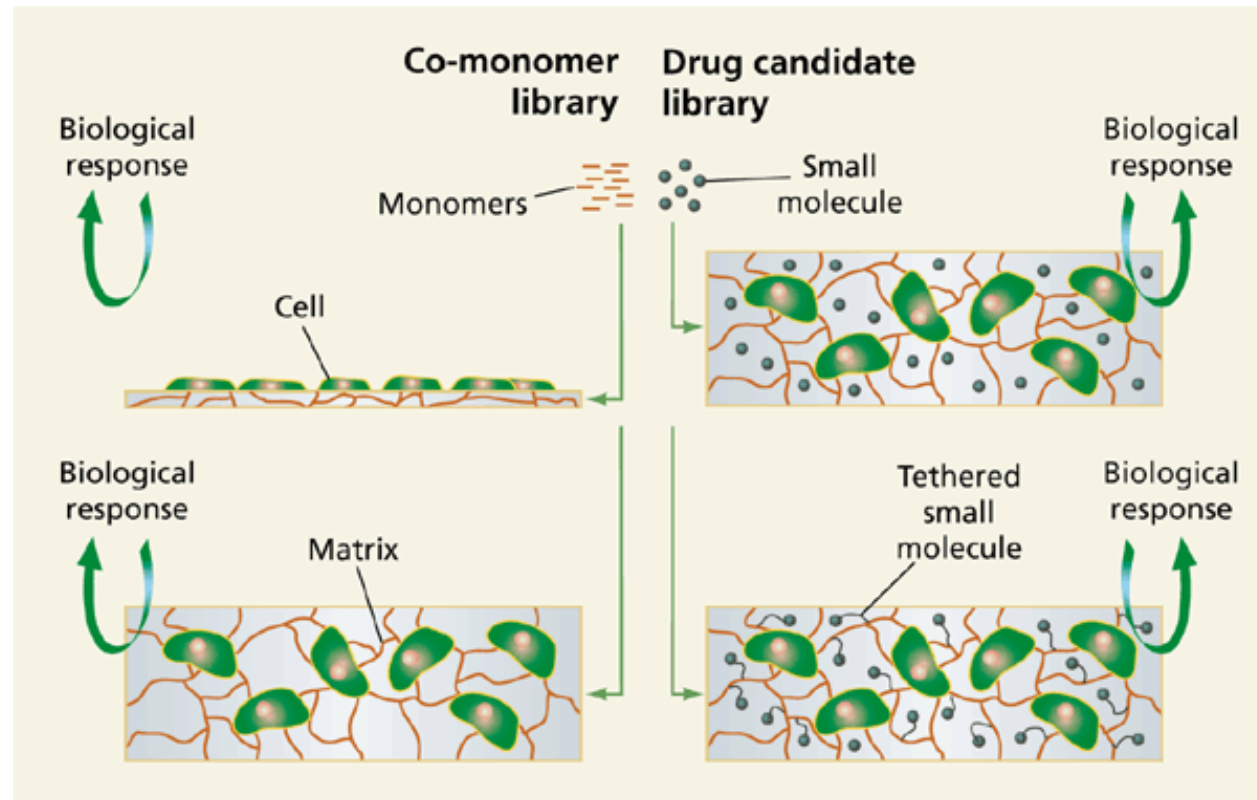
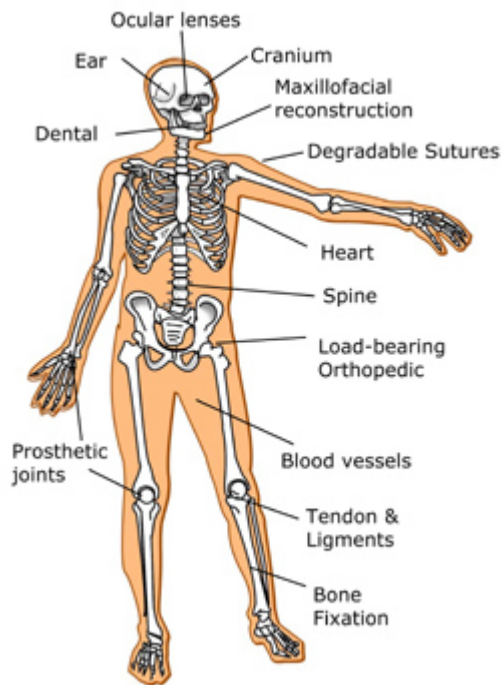


Manufacture of a Medical Device

1. materials selection;
2. types of tests required for evaluation depend on the physical and chemical nature of its materials and the nature of the device's exposure to the body;
3. its physical properties, cost, and availability may be acceptable, but might contain toxic chemical components;
4. screen the candidate materials at an early stage to eliminate those that are toxic, and select those that are sufficiently biocompatible or nontoxic for their intended use.
5. Chemical constituents and potential extractables should be identified and quantitated for overall safety assessment of the device.

Biomaterials Science: an Interdisciplinary Affair

- Bio-materialists include physical scientists, engineers, dentists, biological scientists, surgeons, and veterinary practitioners in industry, government, clinical specialties, and academic settings.



The Society For Biomaterials

A professional society which promotes advances in all phases of materials research and development by encouragement of cooperative educational programs, clinical applications, and professional standards in the biomaterials field. Internationally recognized leaders in the biomaterials field participate in the Society and sponsored events. (www.biomaterials.org)



Relevant Biomaterials Journals

Journal of Biomedical Materials Research

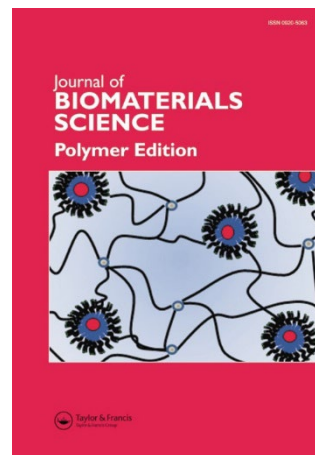
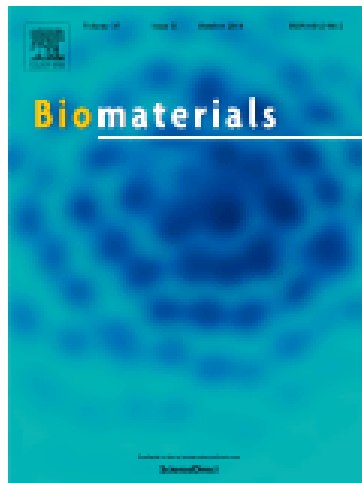
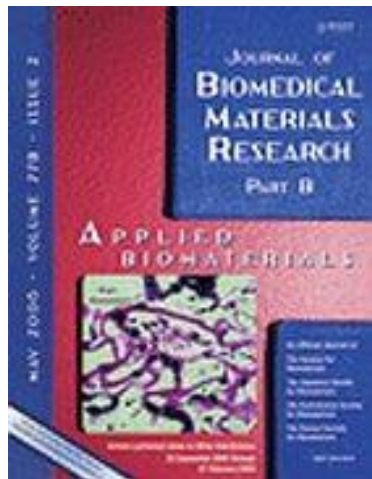
Biomaterials

Journal of Biomaterials Science. Polymer Edition

Journal of Biomaterials Applications

Journal of Materials Science: Materials in Medicine

Advanced Health Materials, Nature Materials, JACS,



Relevant Websites

- ❑ Biomaterials Network (www.biomat.net)
- ❑ Medical Device Information
(www.devicelink.com)
- ❑ Medical Materials Engineering reference
(www.engineeringreference.com)
- ❑ United States Patents and Trademarks Office
(www.uspto.gov)
- ❑ General search-Google (www.google.com)
- ❑ MEDLINE-(www.ncbi.nih.gov/entrez/query.fcgi)

References

Dr. Sudhir Khetan

1. <http://orzo.union.edu/~khetans>
2. <http://orzo.union.edu/~khetans/Teaching/BNG331/L1%20-%20Introduction%20to%20Biomaterials.pdf>
3. <https://www.union.edu/search/?q=biomaterials+course#gsc.tab=0&gsc.q=biomaterials%20course&gsc.page=4>

Dr. Patrick Tresco

1. <http://www.bioen.utah.edu/faculty/pat/Courses/biomaterials/index.html>Amorphous rubber.
2. <http://www.bioen.utah.edu/faculty/pat/Courses/biomaterials/coursenotes.html>

Bulk and surface properties of materials

Bulk Materials are described by:

- Chemical / Molecular composition
- Atomic / Molecular structure (Crystallinity, etc)
- Mechanics (Elasticity, etc)
- Shape

Surfaces of materials have unique descriptive properties:

- Excess surface free energy
- Atomic / Molecular composition
- Chemical composition (reactivity)
- Topography (vs. shape)

Surface characterization provides surface specific information about these properties.

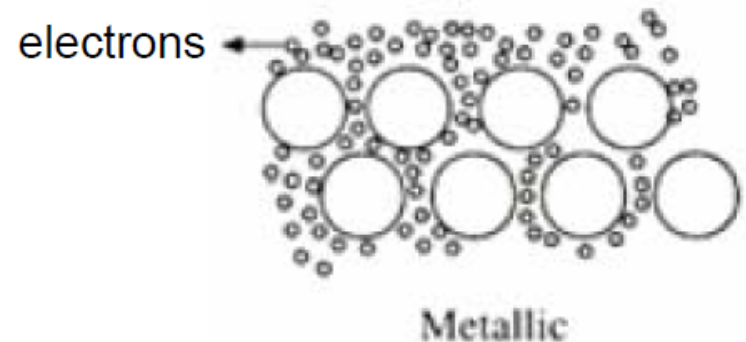
Bulk Properties

The electronic and atomic structures, and almost all the physical properties, of solids depend on the nature and strength of the inter-atomic bonds:

- Ionic Bonding
- Covalent Bonding
- Metallic Bonding
- Weak Bonding – van der Waals and hydrogen bonding

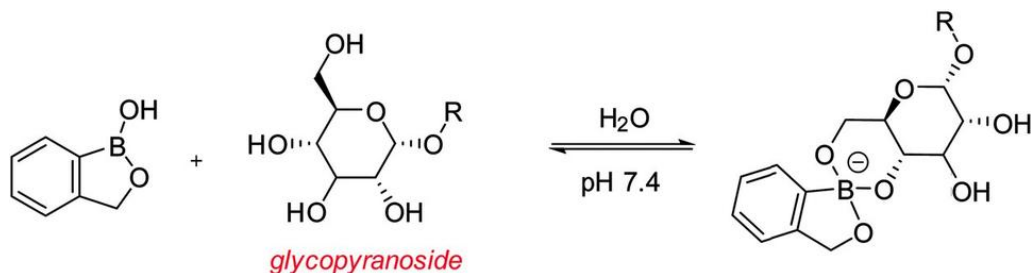
Metallic bonding

- Positively charged ion cores with negative electrons circulate around the core.
 - Free valence electrons which can "travel" about the material.
 - Electrical charge is neutralized on average.
 - Bonds are non-directional (isotropic).
- Allow plastic deformation
 - Atom position is non discriminative
- Good electrical and heat conductance



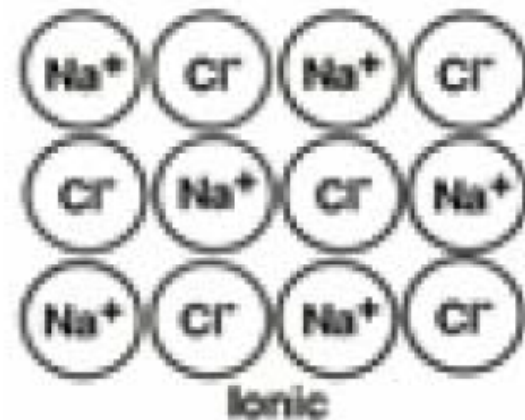
Covalent bonding

- Formed between non-metallic atoms when the valence electrons are shared to fill each outer valence orbital.
 - Carbon atoms sharing pairs of electrons.
 - Highly directional, allowing chains of monomers to be formed.
- Poor electrical and heat conductance.



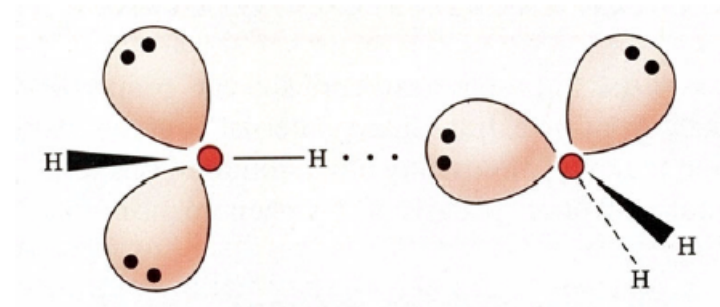
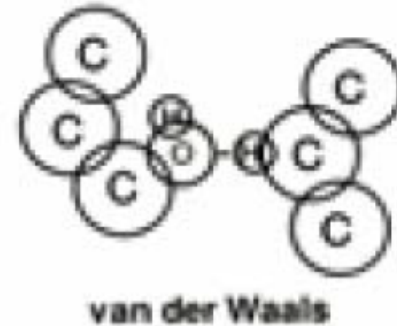
Ionic bonding

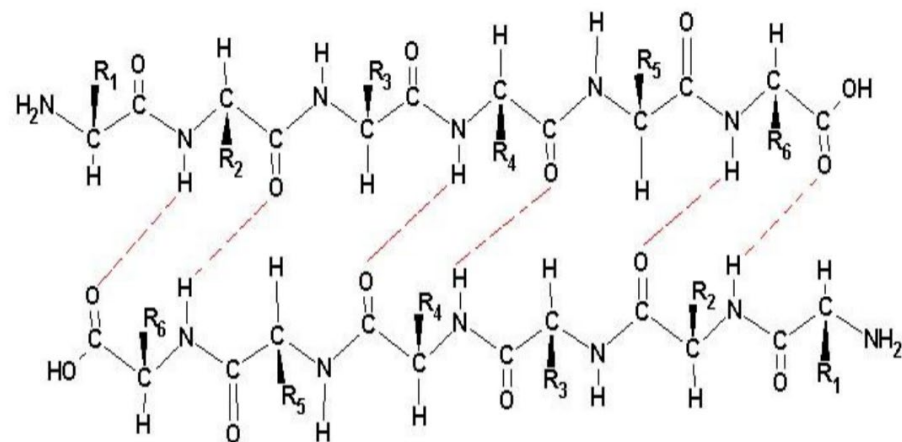
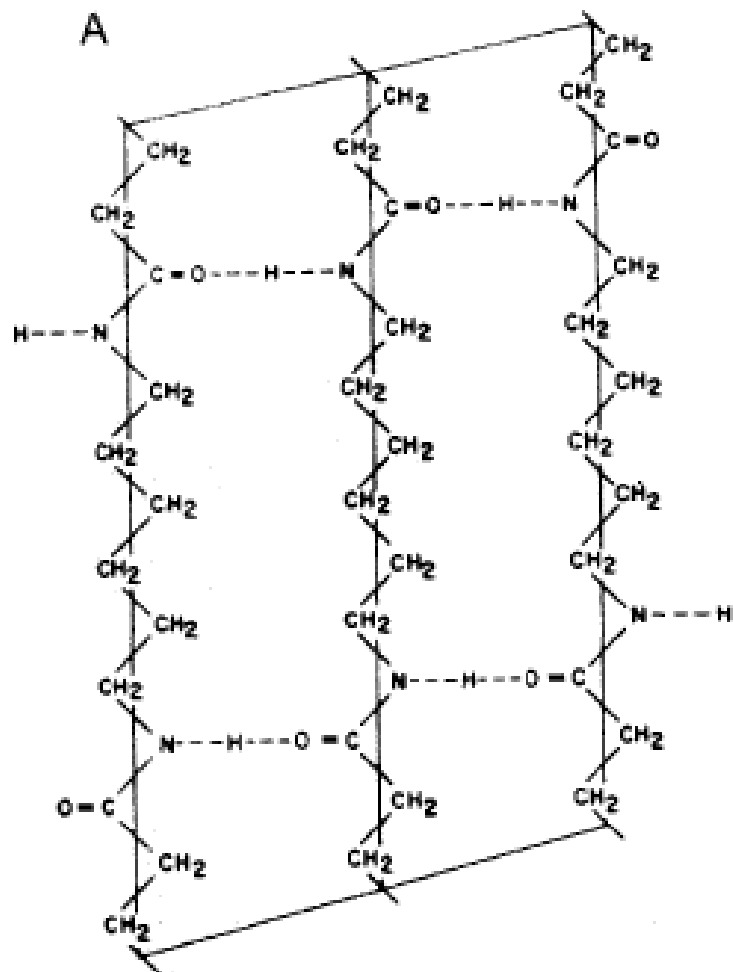
- Metallic atom shares one or more electrons with the non-metallic atom.
 - Cation is surrounded by as many anions as possible.
 - Reduces mutual repulsion of cations.
 - Packing is a function of relative atomic sizes.
- Forms a crystalline structure.
- Results in low electrical and heat conductivity and low chemical reactivity.



Weak or secondary bonding

- Van Waals bonds:
 - Form among hydrogen, oxygen and other atoms.
 - Not highly directional.
 - High number of possible bonds results in strength.
- Hydrogen bonds Hydrogen bonds:
 - Formed when hydrogen is covalently bonded to an electronegative atoms.





蛋白质分子中的氢键示意图

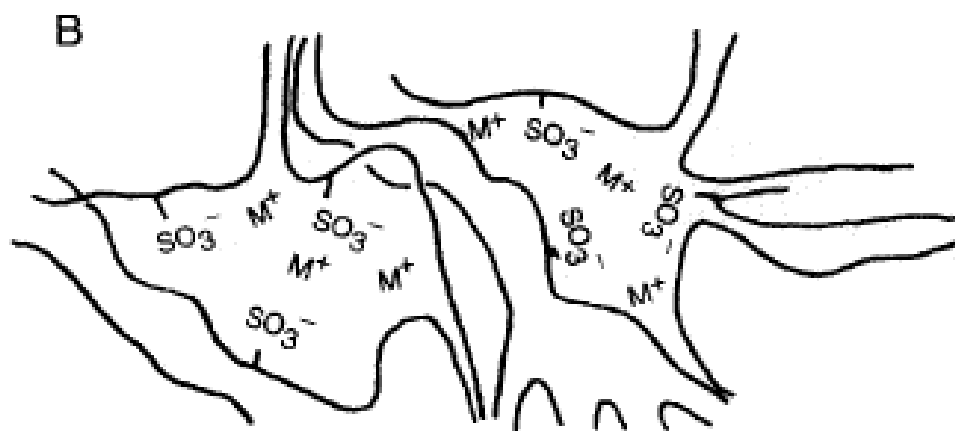


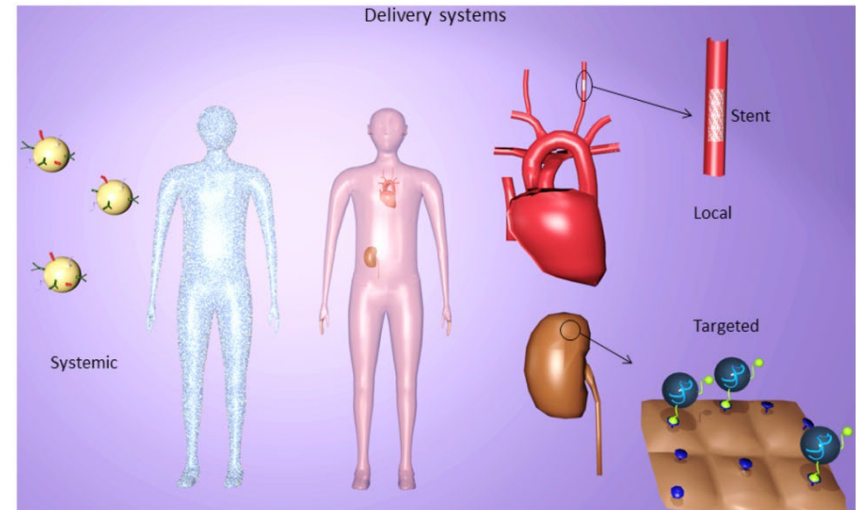
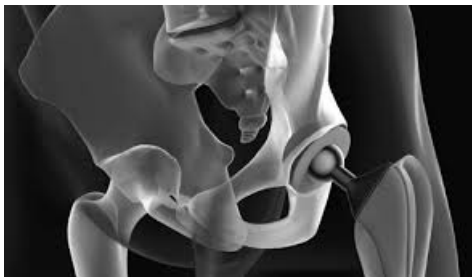
FIG. 4. (A) Hydrogen bonding in nylon 6,6 molecules in a triclinic unit cell: σ -form. (From L. Mandelkern, *An Introduction to Macromolecules*, Springer-Verlag, 1983, p. 43, with permission.) (B) Ionic aggregation giving rise to physical cross-links in ionomers.

Mechanical Properties

Orthopedic applications

Cardiovascular applications

The mechanical properties of the engineered tissues should match that of the host tissue.



Wolff's law

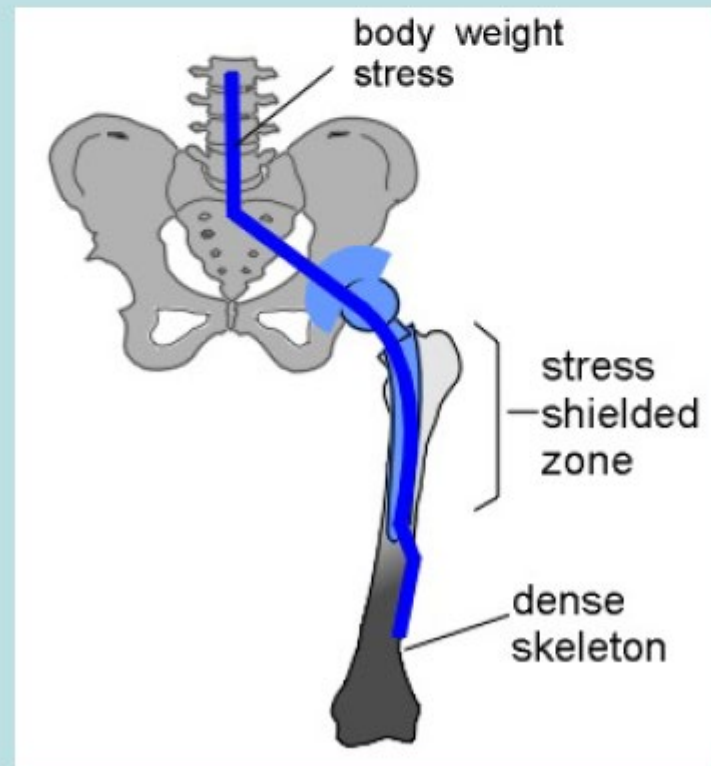
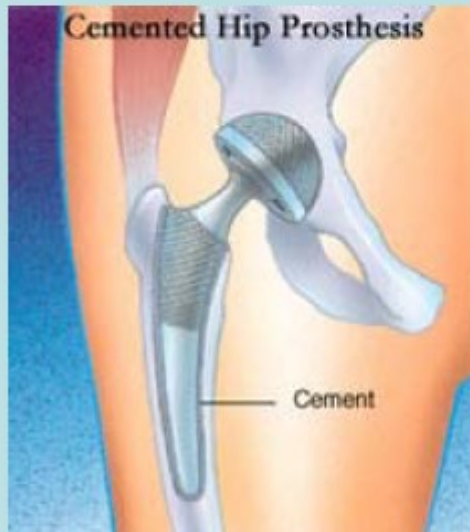
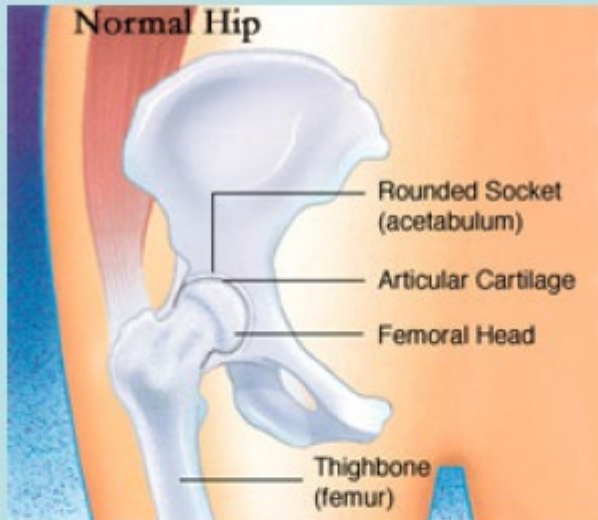
Bone in a healthy person is capable of adapting loads that is placed under.

If loading on a particular bone increases, the bone will remodel itself over time to become stronger to resist the loading.

Conversely, if the loading on a bone decreases, the bone will become weaker.

Stress Shielding (应力遮挡)

The metal alloy implants have a much higher stiffness than the bone. This causes an effect called stress shielding, where all load is transferred through the metal and not the bone, causing the body to resorb the bone.



1. 因内、外固定等方法的应用阻挡或遮断了生理性应力在骨折端的传导从而导致骨连接和骨结构塑型障碍具体讲是对压应力和张应力等有利应力的遮挡所产生的不利效应而对其它有害应力如剪切和旋转应力的克服不能称为应力遮挡
2. 假体的植入将承载一定比例的负荷引起剩余骨质的部分区域应力水平出现下降这种现象被称为应力遮挡.而基于Wolff定律骨质改建定律说明骨质将随着机械力学微环境的变化作出股骨近端骨质的改进在应力遮挡区域即出现一定的骨吸收
3. 这种现象称为应力遮挡.它将引起骨吸收,这是影响人工关节置换远期疗效的主要并发症之一.加之金属材料属于生物惰性材料,生物活性较差,植入人体后,始终作为宿主的异物存在,容易变形、松动
4. 接骨器的应力遮挡效应所谓应力遮挡是指骨折用器械固定后,压应力刺激被器械遮挡了.这种浓度概念并不能准确反映实际施加到织物上的量
5. 所谓的应力遮挡是指股骨近端应力在术后由假体承担了大部分的应力,造成相应区域骨质应力的降低.脊柱失稳是指脊柱的这种能力丧失而导致椎体移位超出限度的病理过程
6. 其原因被认为是应力保护(常被误称为应力遮挡)和钢板下皮质骨因血供破坏而致哈佛系统加速重塑.而一旦细胞的这种生与死失去平衡,人体就会产生很多疾病
7. 因而会引起周围的骨组织弱化,这种现象称为应力遮挡.应力遮挡效应将引起骨被吸收[2].这是导致人工骨节置换远期并发症之一,同时也是造成植入人工金属假体的无菌性松动、下沉以及有害组织重塑的重要原因

“Diminished **shield effect**” of the femoral shaft component is one of the selling argument of new models of **total hip devices**. But can one really produce artificial total hips with **stiffness** values that are almost identical with the stiffness of the skeleton around the total hip device?

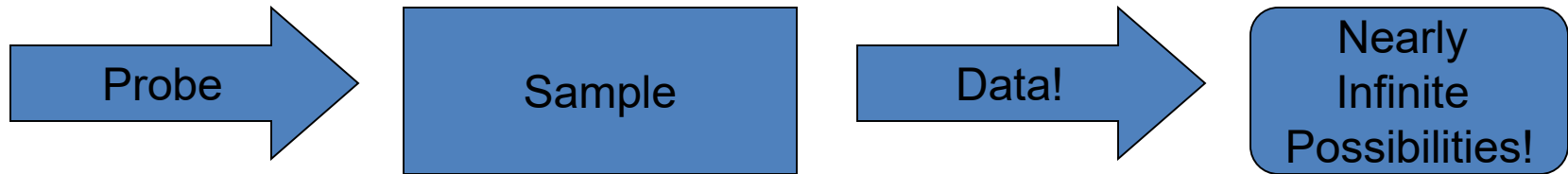
With **Tissue Engineering**, you can!

Characterization of Mechanical Properties

Characterization of mechanical properties of materials is important for matching biomaterial properties to the *in vivo* microenvironment.

- Elasticity – typical reported parameter:
- Elastic modulus (slope of stress-strain curve)
- Yield strength (the stress at which a given amount of plastic deformation occurs)
- Ultimate tensile strength (the stress at which the material fails)
- Ductility (the total plastic strain exhibited before fracture)
- Toughness (area under the stress-strain curve until failure)

Characterization



Characterization is the method by which one develops a data set that describes properties of the sample. Because of limited possibilities this process is:

- Discrete / Reliant on methodology
- Application specific
- Often material specific
- Resource limited

Surface Sensitivity

The further your technique returns info from the sample, the less sensitive it is to the surface. This is usually relative to the penetration depth of the probe.

Attenuated total reflection (ATR)

减弱全反射

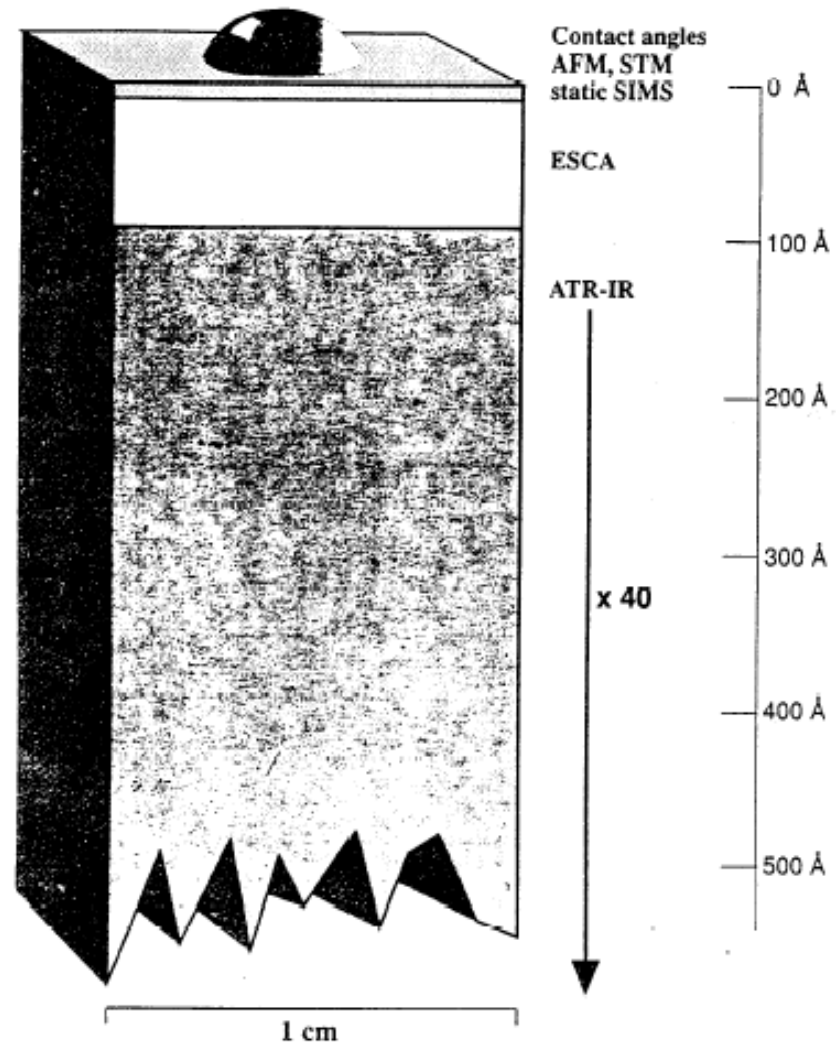


Figure 2. The major surface analysis methods probe to different depths. Contact angles, scanning tunneling microscopy, atomic force microscopy, and static secondary ion mass spectrometry are extremely surface localized.

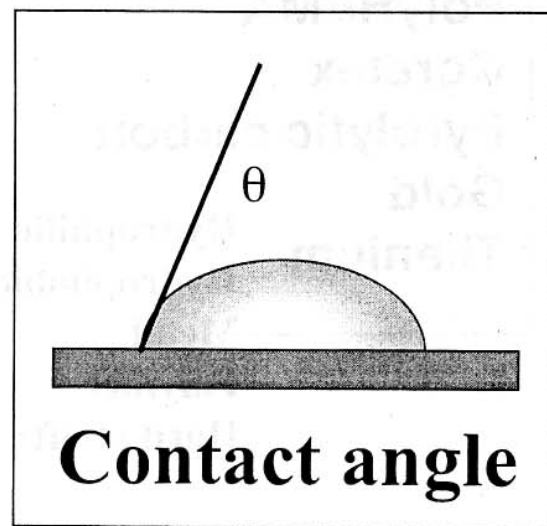
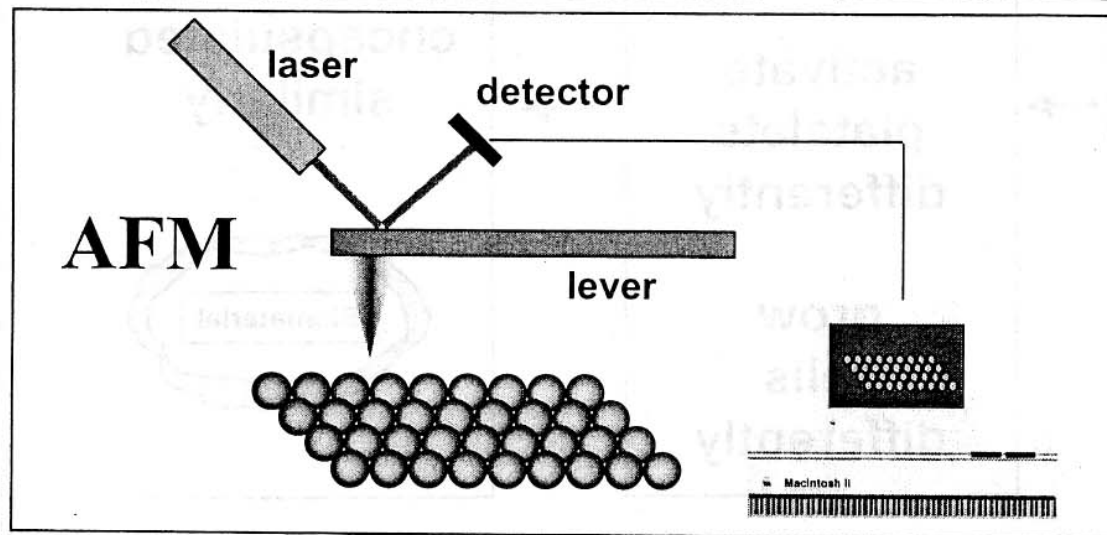
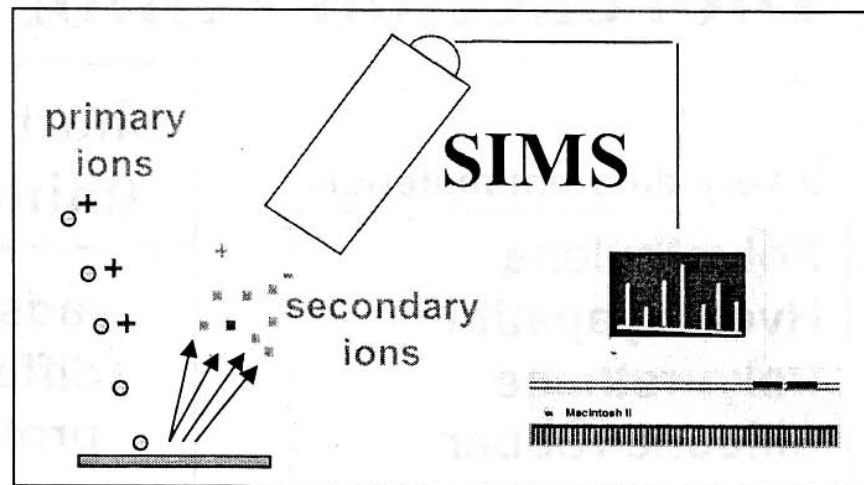
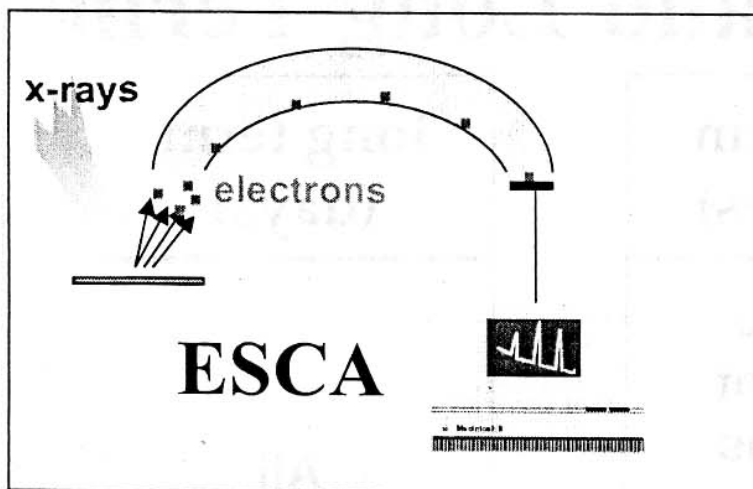
Surface Analysis Techniques for Biomaterials

- Contact Angle Measurements
- Electron Spectroscopy for Chemical Analysis (ESCA / XPS)
- Auger Electron Spectroscopy (Auger) (俄歇电子能谱)
- Near Edge X-ray Absorption Fine Structure (NEXAFS)
(近边x射线吸收精细结构)
- Secondary Ion Mass Spectroscopy (SIMS)
- Scanning Probe Microscopy (AFM)
- Sum Frequency Generation (SFG) (和频振荡)
- Surface Plasmon Resonance (SPR)
- Optical Imaging and Spectroscopy (microscopy, TIRF)
- Ellipsometry (椭圆偏振技术)
- Scanning Electron Microscopy (SEM)
- Infrared Spectroscopy (FTIR)
- Many more...

Surface Information

<u>Property</u>	<u>Technique(s)</u>
Composition	ESCA, Auger, SIMS, NEXAFS
Structure	SIMS, ESCA, NEXAFS, FTIR, SFG
Orientation	NEXAFS, FTIR, SFG
Spatial Distribution	Imaging SIMS, AFM, microscopy
Topography	AFM
Thickness	ESCA, AFM, ellipsometry, SPR
Energetics	Contact angle

The Basic Repertoire



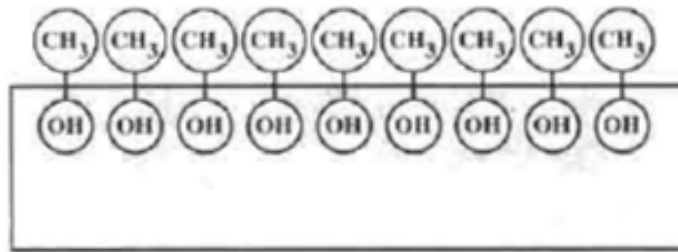
Contact Angle Technique

Modes:	Sessile drop, captive bubble, Wilhelmy, dynamic
Probe:	Small Drop of Liquid or Bubble
Data:	Contact Angle (Θ)
Sample:	Any material interface that can support the probe
Principle:	Interfacial tension can be used to estimate solid surface energetics
Information:	Surface energetics
Depth:	Å's
Spatial Resolution:	mm ²
Sensitivity:	Depends on chemistry
Relative Cost:	Inexpensive
Other:	Similar techniques can be used for liquids Can be used to estimate solid surface energies

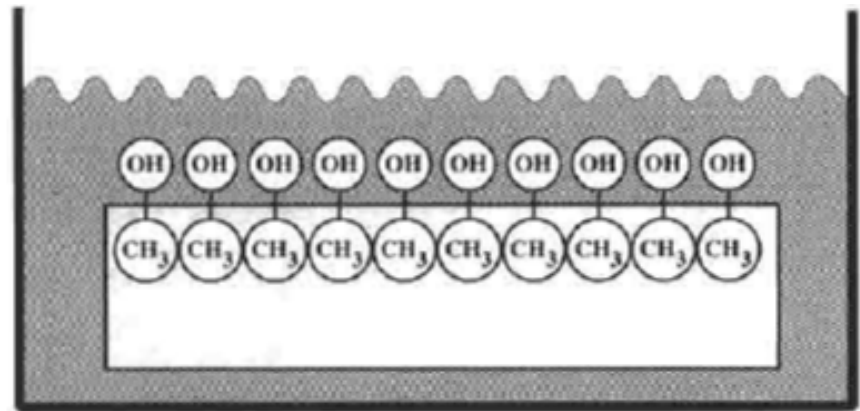
Contact Angle Instrument



Many materials under a reversal of surface structure when transferred from air into a water environment. In this schematic illustration, a hydroxylated polymer exhibits a surface rich in methyl groups.



In air

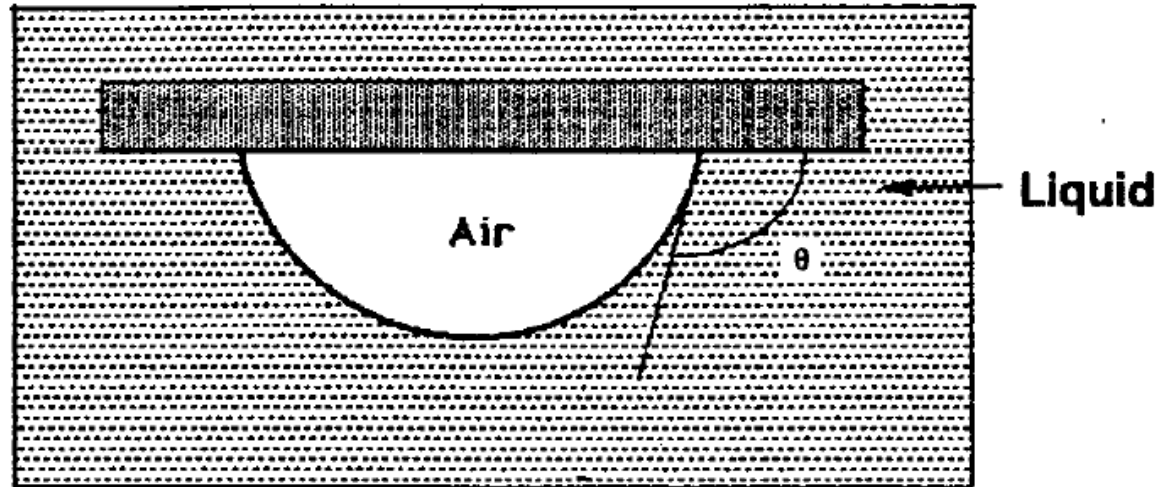


Under water

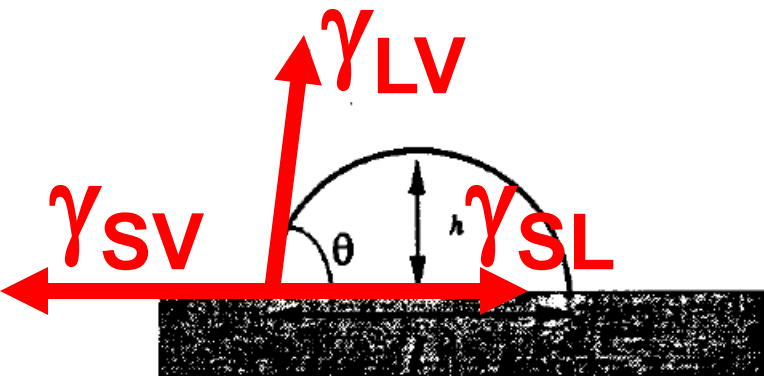
After class, please find one (there are many)

Equilibrium Method

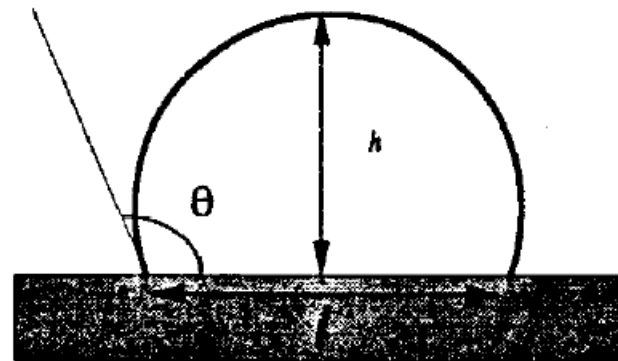
(a) Captive Bubble Method



(b) Sessile Drop for $\theta < 90^\circ$

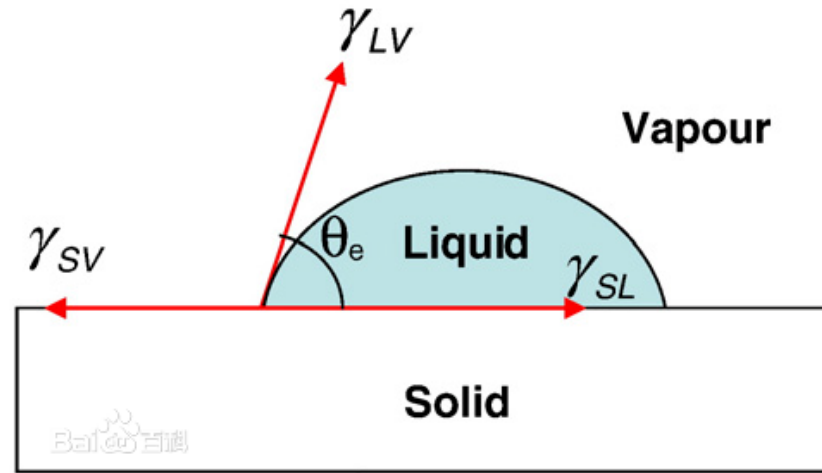


(c) Sessile Drop for $\theta > 90^\circ$



Young equation

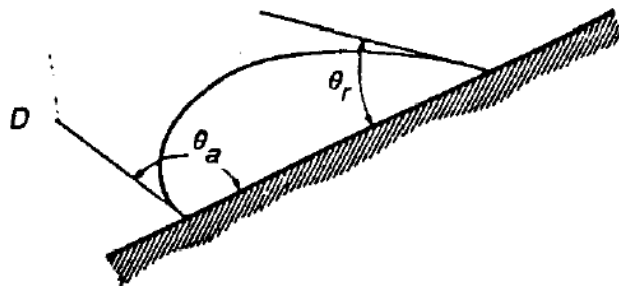
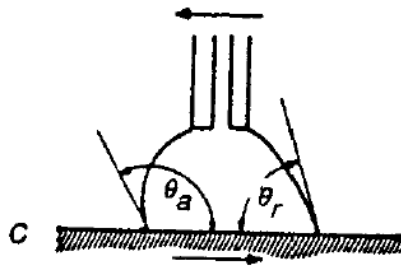
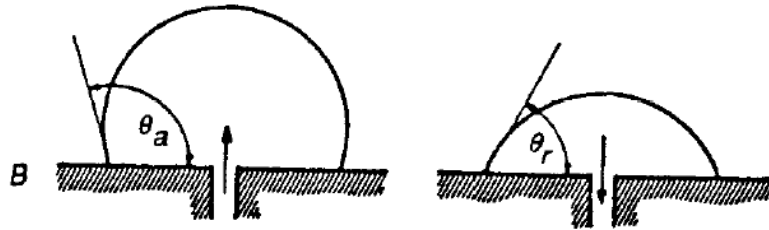
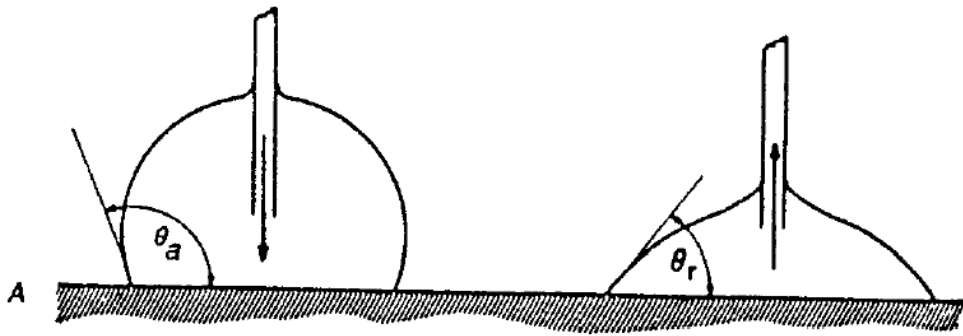
Young contact angle θ_Y



$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \times \cos\theta_e$$

- 1) 当 $\theta=0$ ，完全润湿；
- 2) 当 $\theta < 90^\circ$ ，部分润湿或润湿；
- 3) 当 $\theta=90^\circ$ ，是润湿与否的分界线；
- 4) 当 $\theta > 90^\circ$ ，不润湿；
- 5) 当 $\theta=180^\circ$ ，完全不润湿。

Dynamic Method

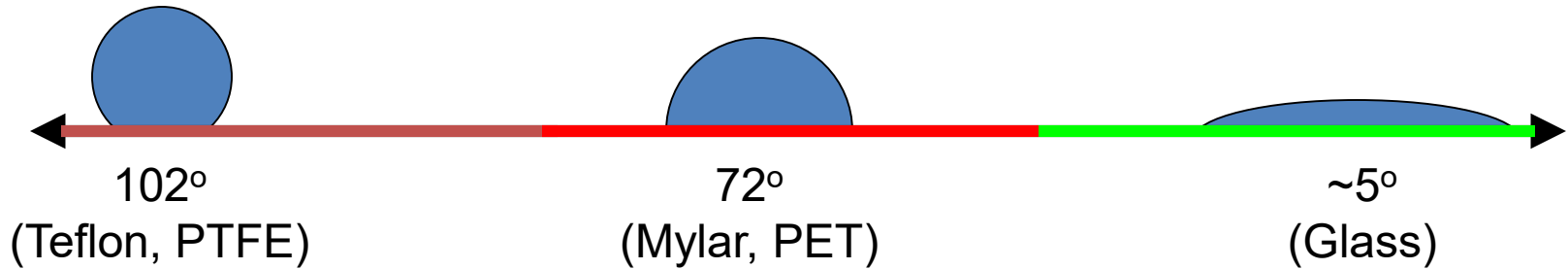


Advancing and receding contact angles give the maximum and minimum values the static contact angle can have on the surface.

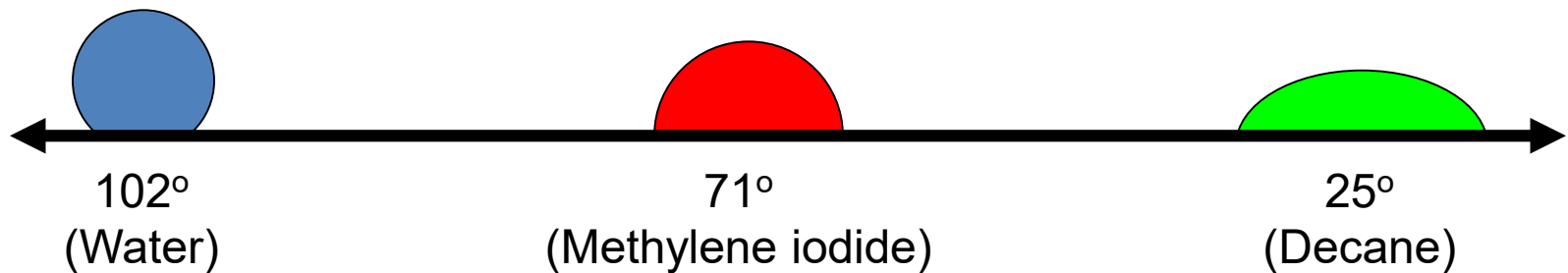
If one observes “hysteresis” or a difference in θ_a vs. θ_r then it is likely that the surface is dynamically rearranging in response to interaction with the probe.

Θ Indicates Degree of Wetting

For water ($\gamma_{LV} = 72.8 \text{ mN/m}$) this is the degree of hydrophobicity and it varies with material:



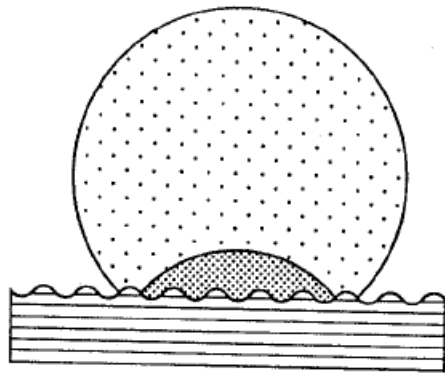
One can also vary the probe, for example if Teflon is used:



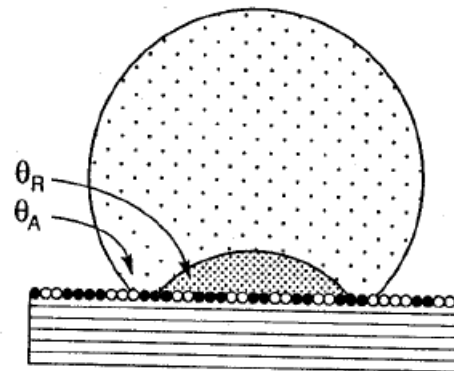
Limitation of Contact Angles

Measurement is highly operator dependent, affected by contamination, affected by surface roughness heterogeneity and rearrangements, and there are limited sample geometries.

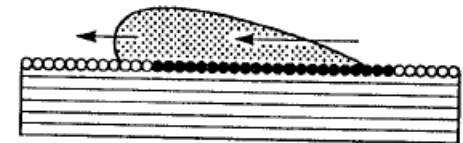
Biggest problem – so cheap and easy it is often done poorly and misinterpreted.



Surface Roughness



Heterogeneity



Rearrangements

Characterization Methods for Biomaterial Surfaces

Method	Principle	Depth Analyzed	Spatial Resolution	Analytical Sensitivity	Cost ^c
Contact angles	Liquid wetting of surfaces is used to estimate the energy of surfaces	3–20 Å	1 mm	Low or high depending on the chemistry	\$
ESCA (XPS)	X-rays induce the emission of electrons of characteristic energy	10–250 Å	10–150 µm	0.1 atom %	\$\$\$
Auger electron spectroscopy ^a	A focused electron beam stimulates the emission of Auger electrons	50–100 Å	100 Å	0.1 atom %	\$\$\$
SIMS	Ion bombardment sputters secondary ions from the surface	10 Å–1 µm ^b	100 Å	Very high	\$\$\$
FTIR-ATR	IR radiation is adsorbed and excites molecular vibrations	1–5 µm	10 µm	1 mol %	\$\$
STM	Measurement of the quantum tunneling current between a metal tip and a conductive surface	5 Å	1 Å	Single atoms	\$\$
SEM	Secondary electron emission induced by a focused electron beam is spatially imaged	5 Å	40 Å, typically	High, but not quantitative	\$\$

^a Auger electron spectroscopy is damaging to organic materials and is best used for inorganics.

^b Static SIMS ≈ 10 Å, dynamic SIMS to 1 µm

^c \$, up to \$5000; \$\$, \$5000–\$100,000; \$\$\$, >\$100,000.

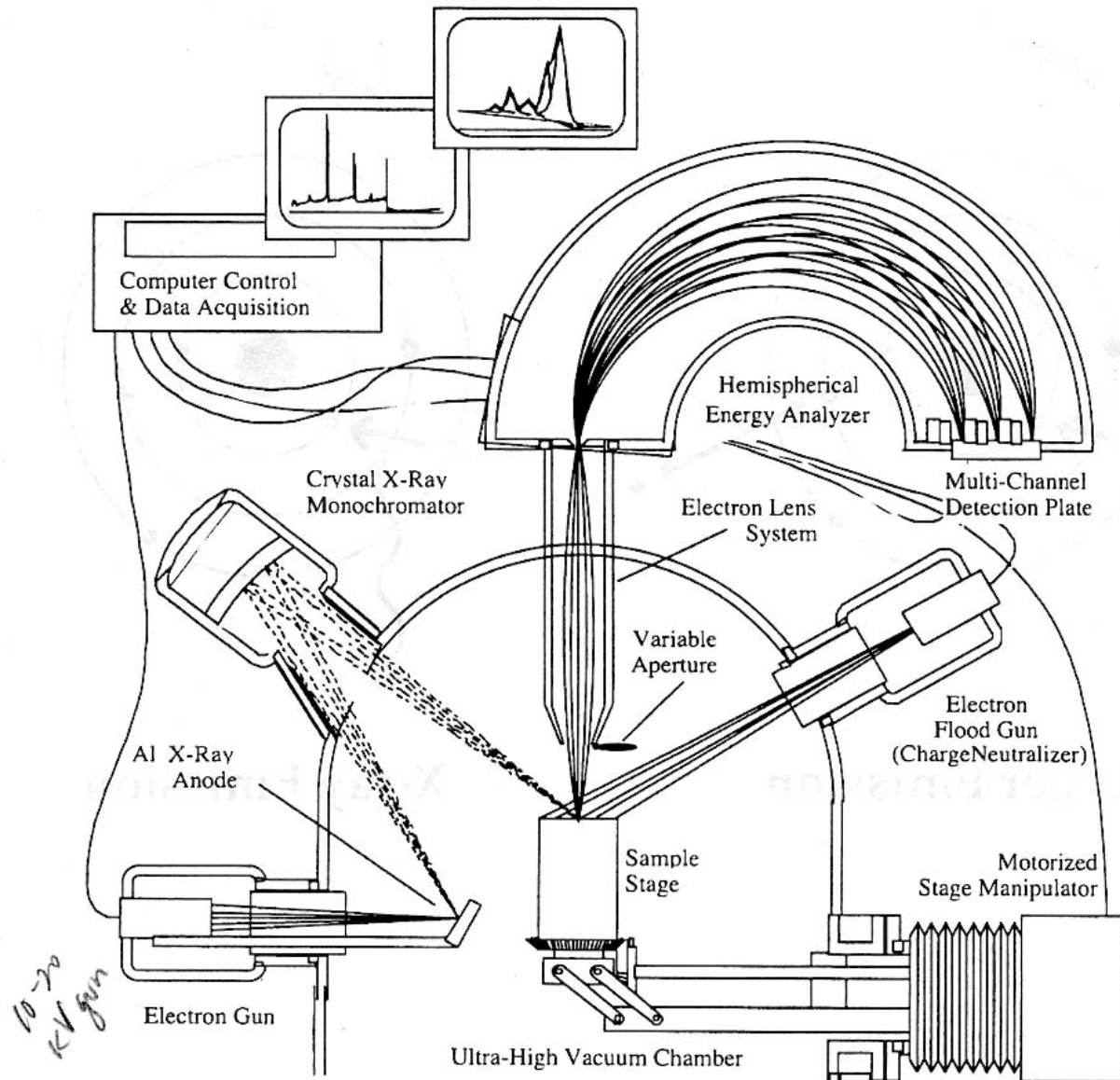
Photoelectric Techniques

Alias:	ESCA (Electron Spectroscopy for Chemical Analysis)
Modes:	XPS (x-ray photoelectron spectroscopy), Auger Spectroscopy, UPS (UV photoelectron spectroscopy)
Probe:	Photons (x-rays, UV)
Signal:	Electrons
Information:	Elemental Composition and molecular environment
Sample:	Any that can withstand ultra-high vacuum
Principle:	Photoelectric effect (think Einstein)
Depth:	100 Å (1000+ Å in destructive mode)
Spatial Resolution:	μm^2
Sensitivity:	1% error not unreasonable
Relative Cost:	Very Expensive
Other:	Semi-quantitative to quantitative NEXAFS is an extension of this technique that gives orientation information Imaging modes

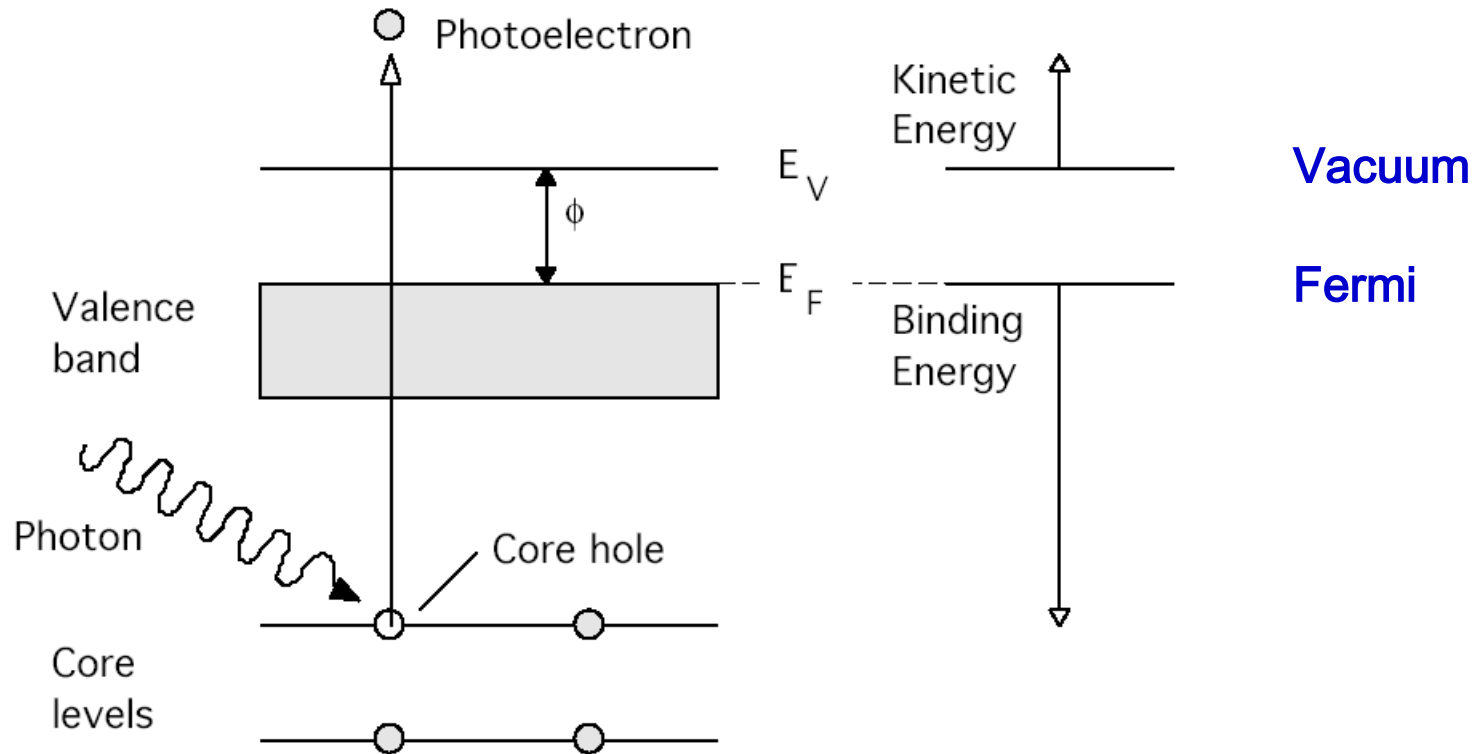
Instrument



Instrument

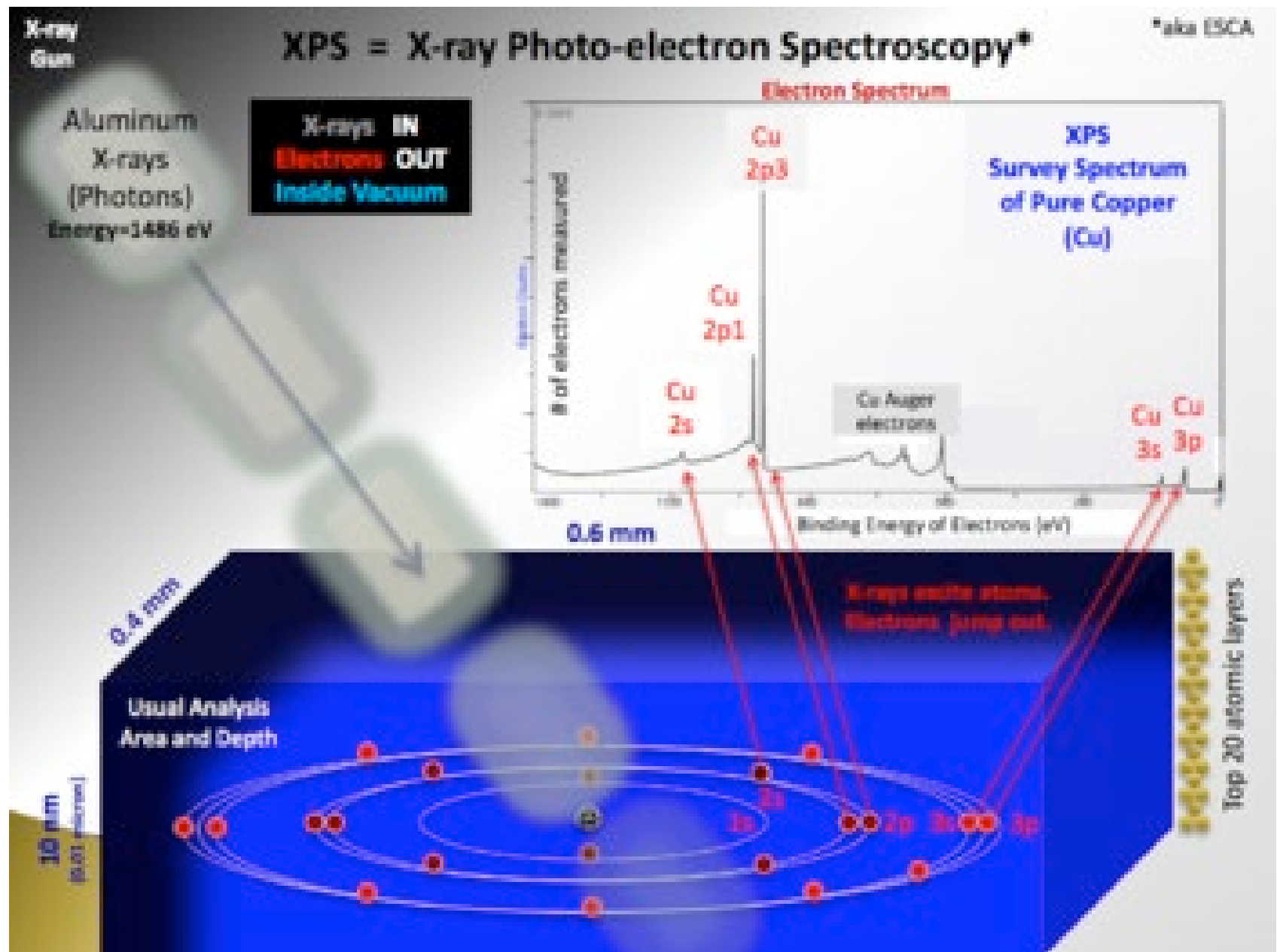


The Photoemission Process



$$KE = h\nu - BE - \phi$$

When an atom or molecule absorbs an X-ray photon, an electron can be ejected. The kinetic energy (KE) of the electron depends upon the photon energy ($h\nu$) and the binding energy (BE) of the electron (i.e., the energy required to remove the electron from the surface).



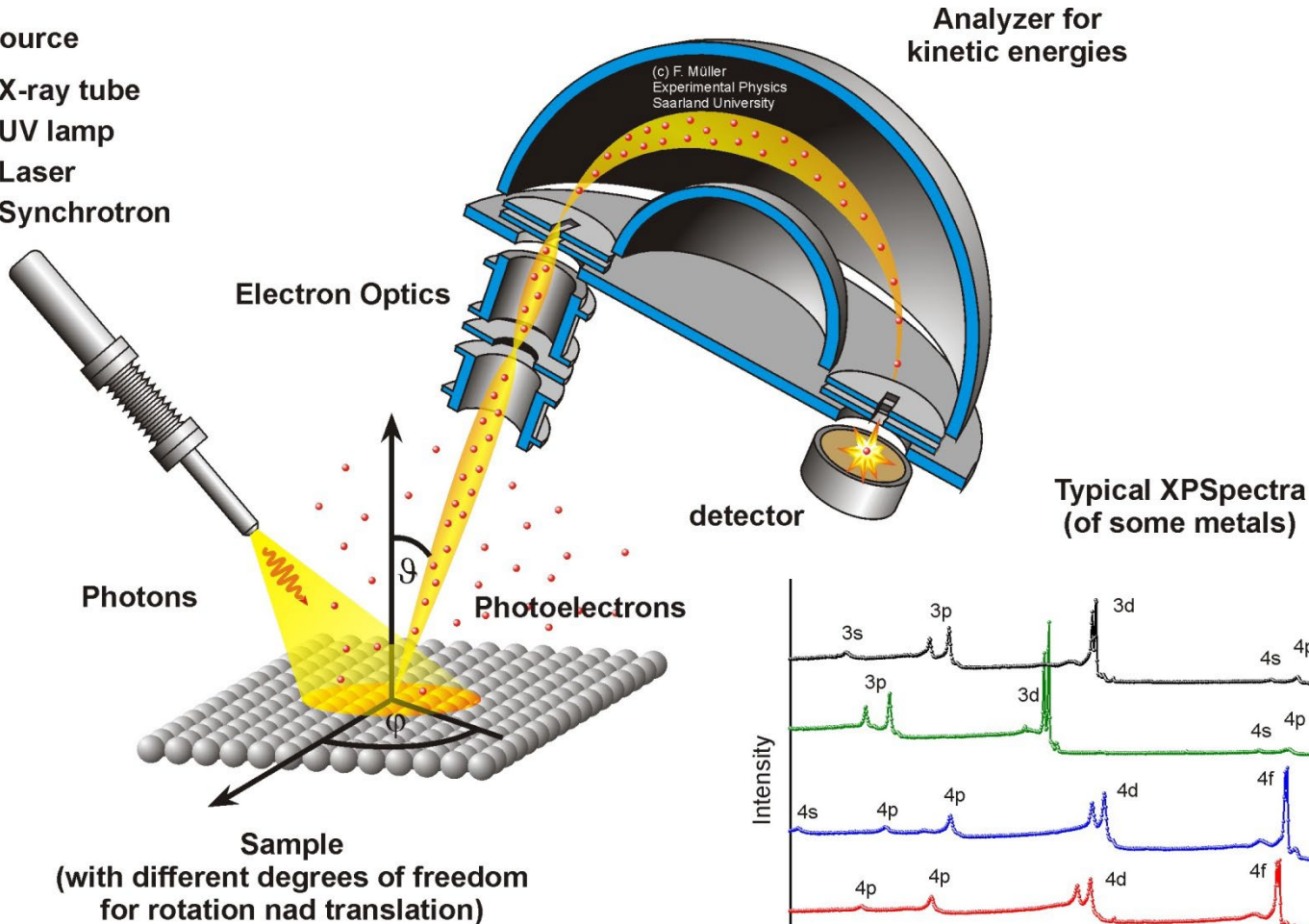
Data from ESCA

Steps due to inelastic background (function of orbitals)

Need to compare to established binding energy curves

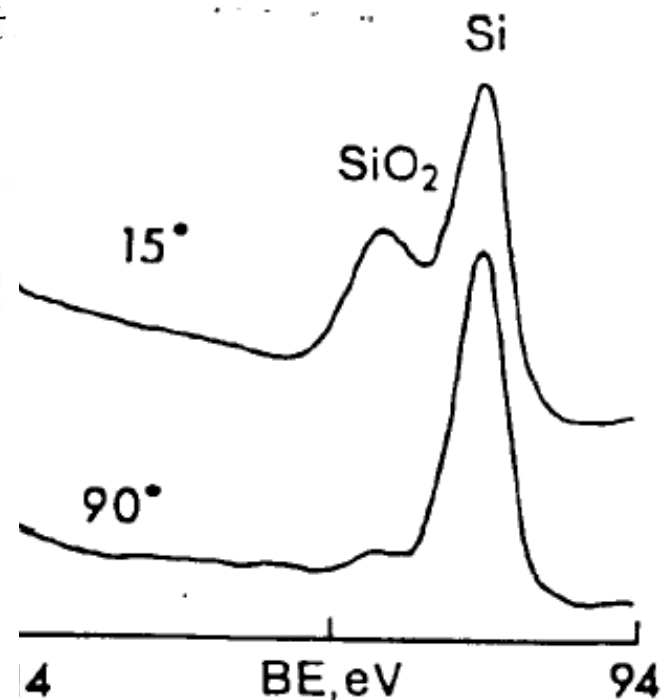
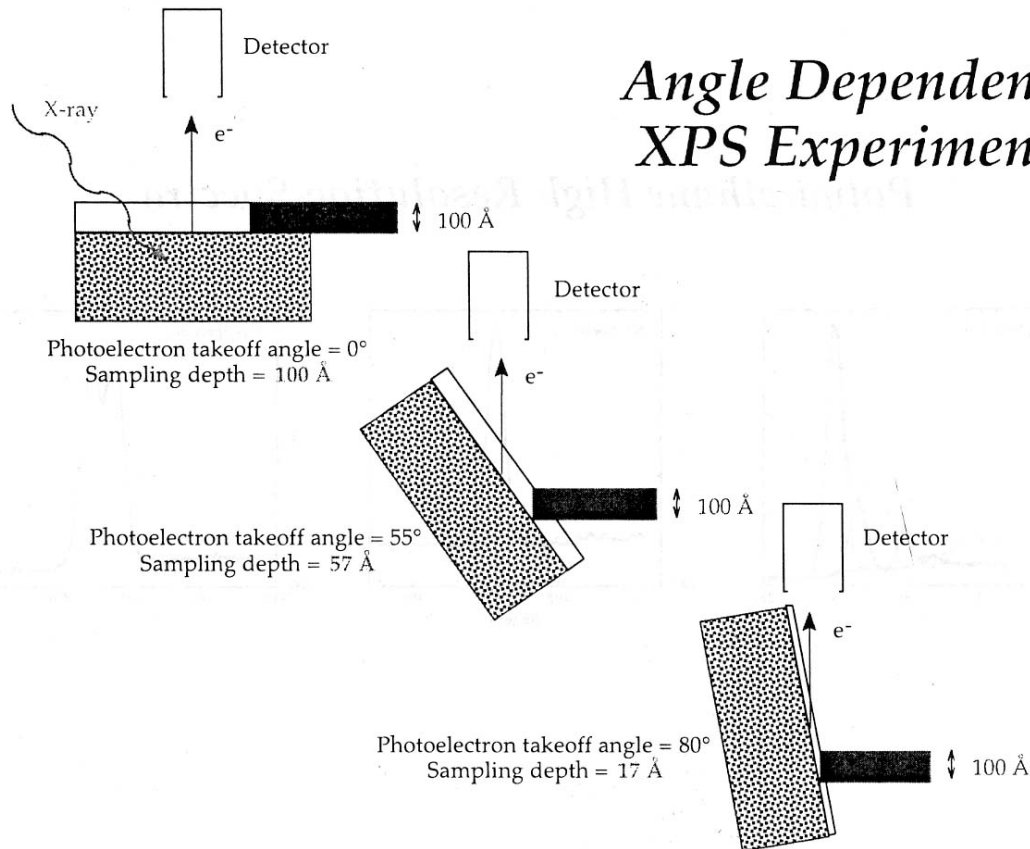
Photon Source

- X-ray tube
- UV lamp
- Laser
- Synchrotron



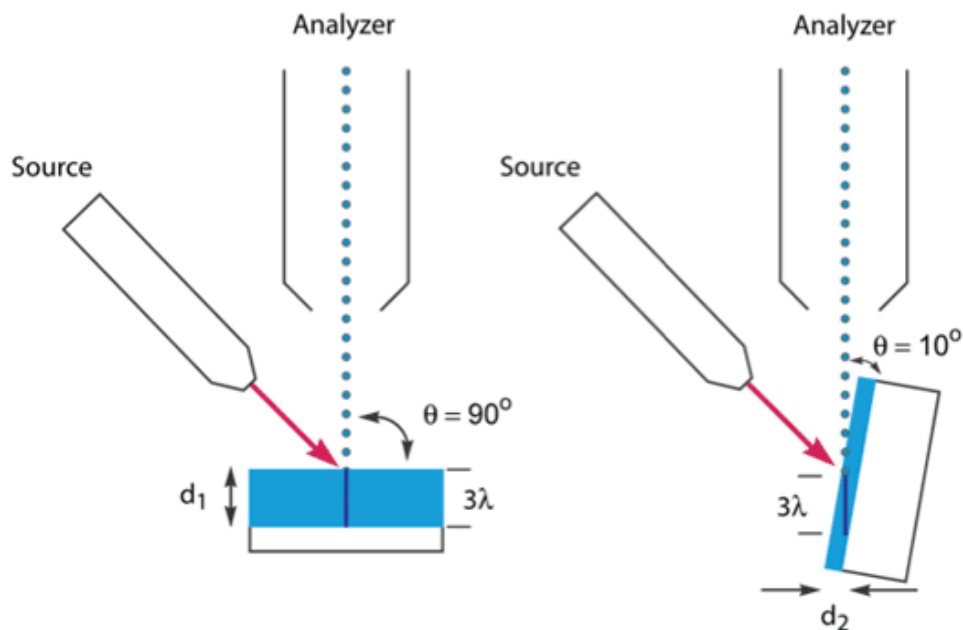
Depth Scanning

ESCA and related techniques can be made more surface sensitive by angling in the beam.

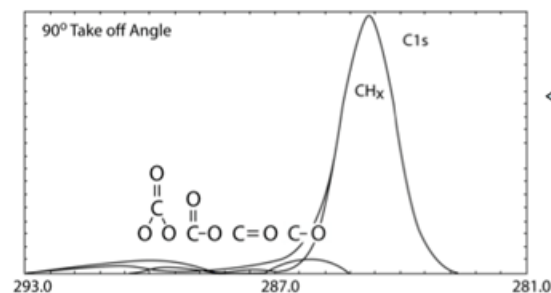


Sampling depth is defined as the depth from which 95% of all photoelectrons are scattered by the time they reach the surface (3λ). Most λ s are in the range of 1 – 3.5 nm for AlK_α radiation.

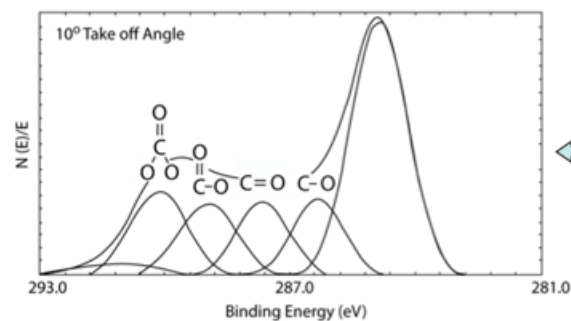
Depth Scanning



$d = \text{Sampling Depth (脱出深さ)} = \sim \lambda \sin \theta$
 λ が一定なので、 $d_2 < d_1$



← 深い情報

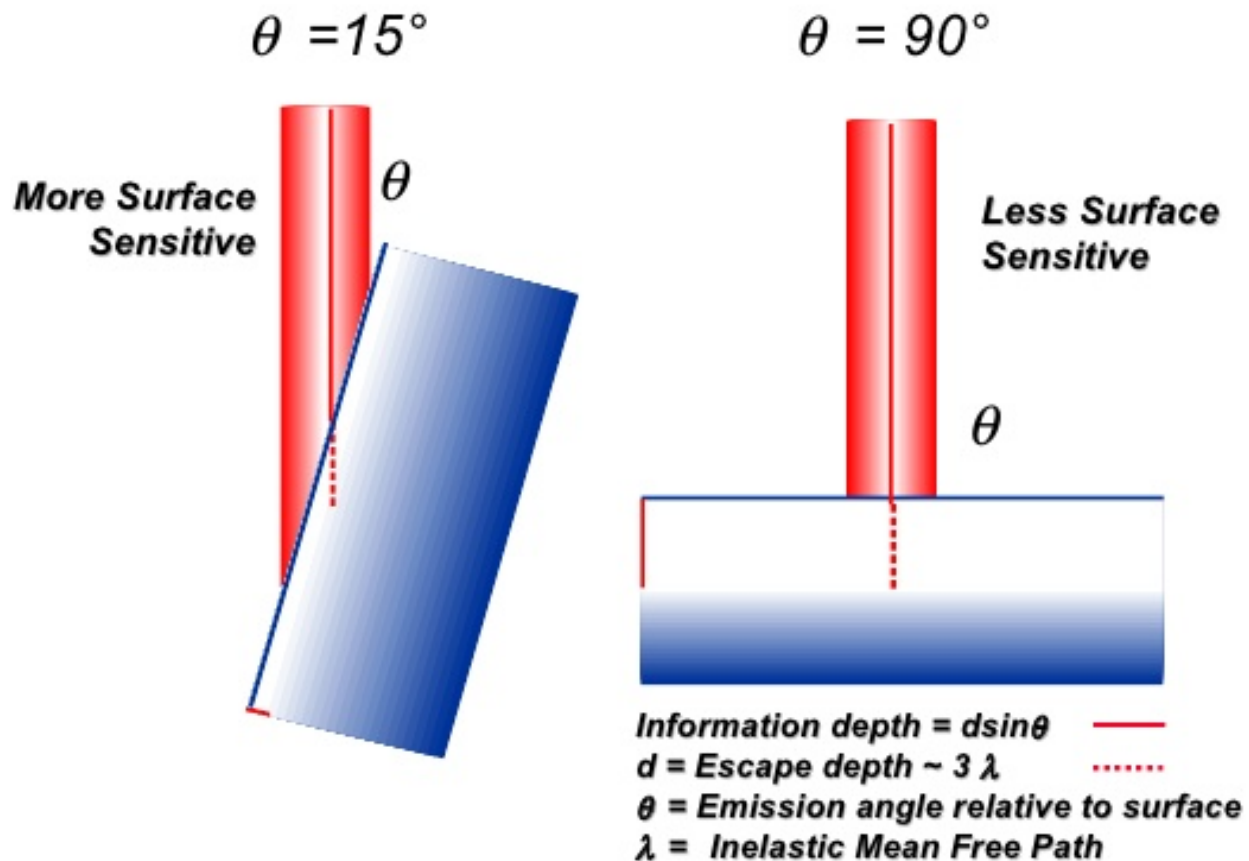


← 浅い情報(極表面付)

<http://www.nanoscience.co.jp/knowledge/XPS/knowledge01.html>

Depth Scanning

Angle-resolved XPS

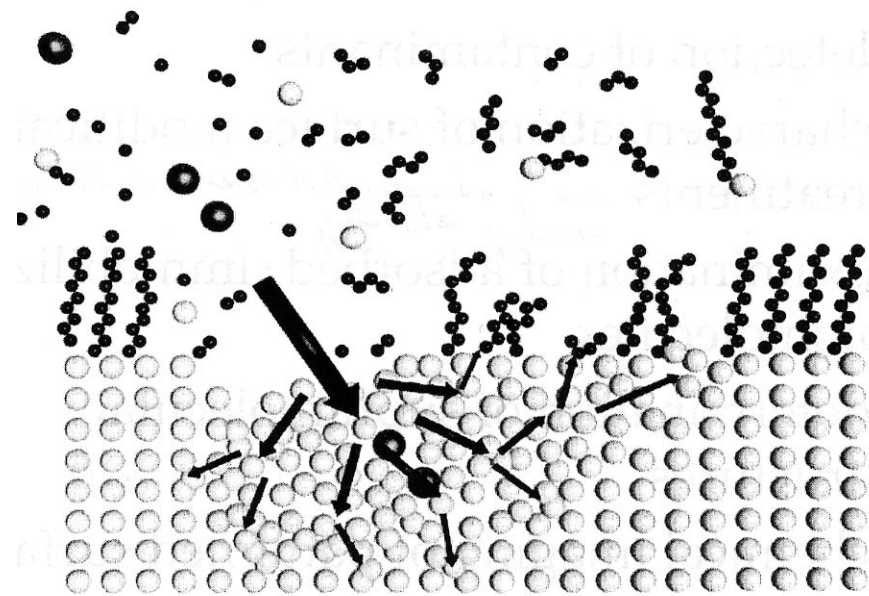
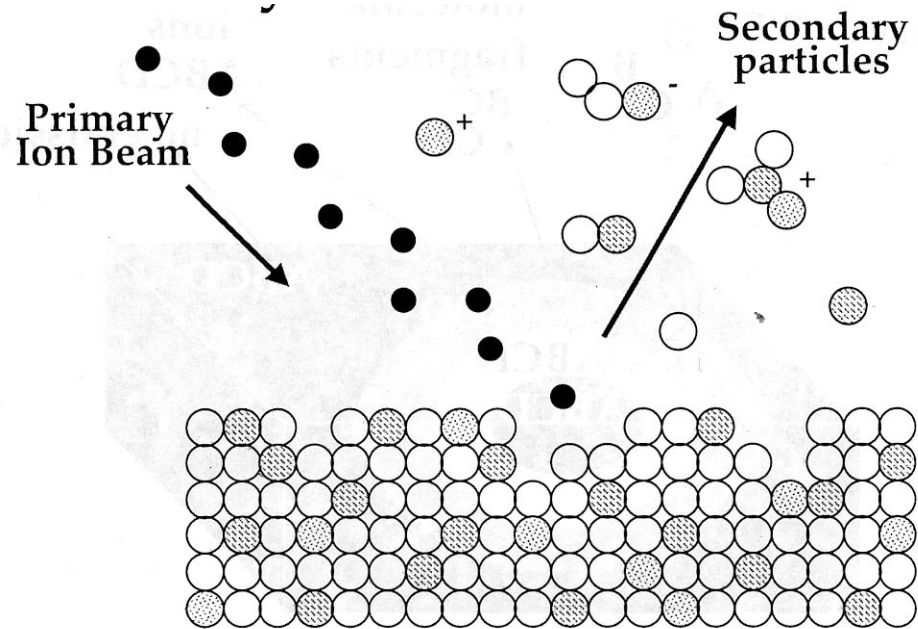


SIMS (次级离子质谱法)

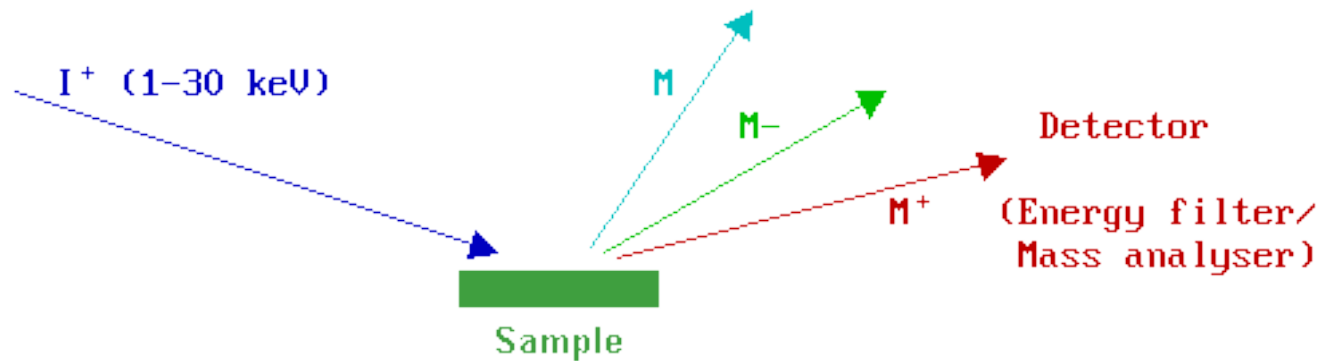
Alias:	Secondary Ion Mass Spectroscopy, ToF-SIMS
Modes:	static and dynamic, secondary electron
Probe:	Ions (Ar^+ , Ga^+ , Cs^+ , C_{60}^+ , etc.), keV
Signal:	Secondary Ions (from sample)
Information:	Elemental and Molecular Composition
Sample:	Any that can withstand ultra-high vacuum
Principle:	Bombardment ions liberate secondary particles from the surface, secondary ions can be detected for mass
Depth:	10 Å (more in dynamic mode)
Spatial Resolution:	less than 1 μm^2
Sensitivity:	“very high”
Relative Cost:	Expensive
Other:	Semi-quantitative to quantitative Can resolve isotopes Imaging modes ToF detectors lead to exact mass detection

SIMS

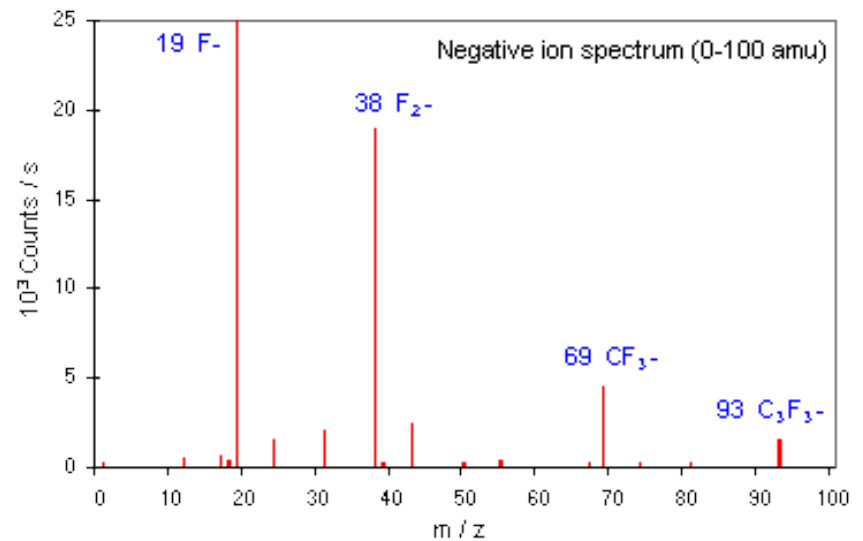
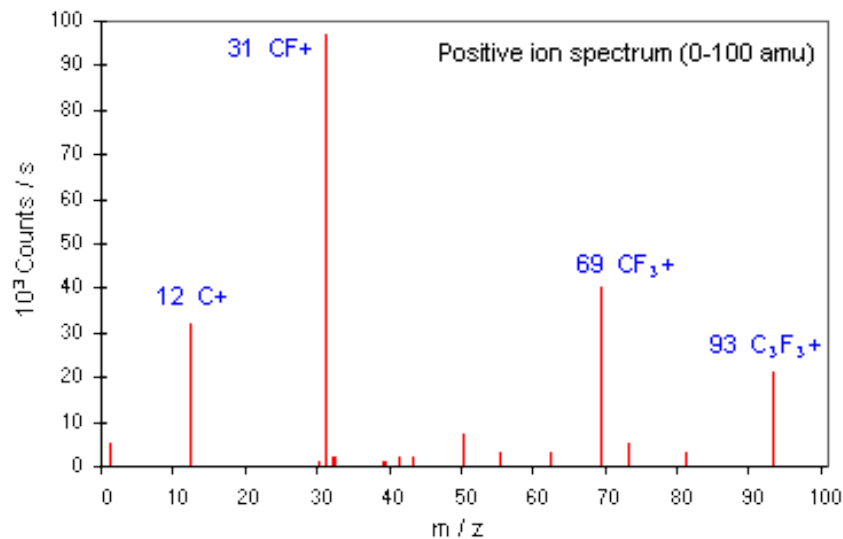
This is a (by design) a destructive technique...



SIMS



Secondary ions can be atomic and molecular, positive or negative (99% of scattered mass is neutral!)



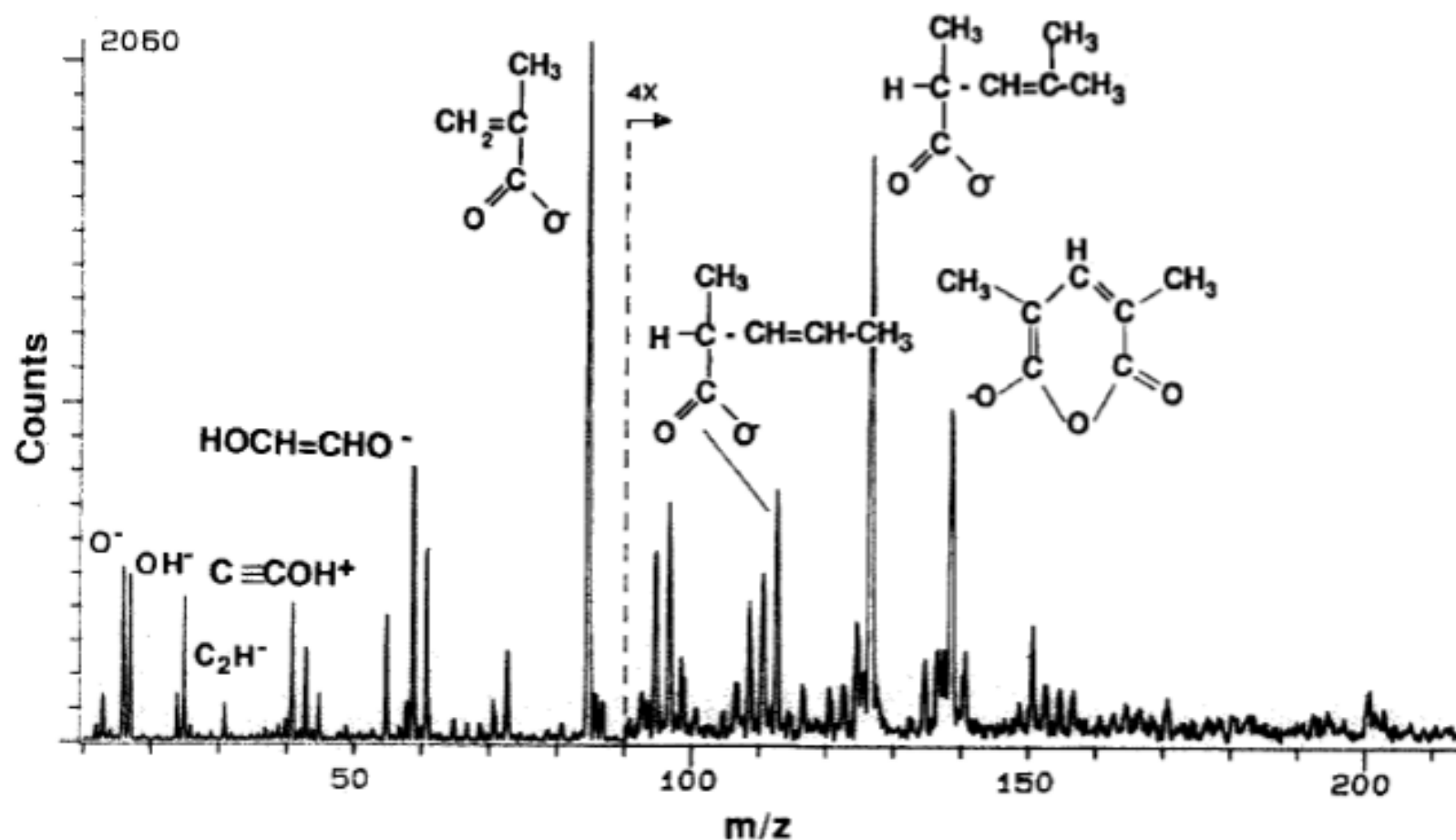


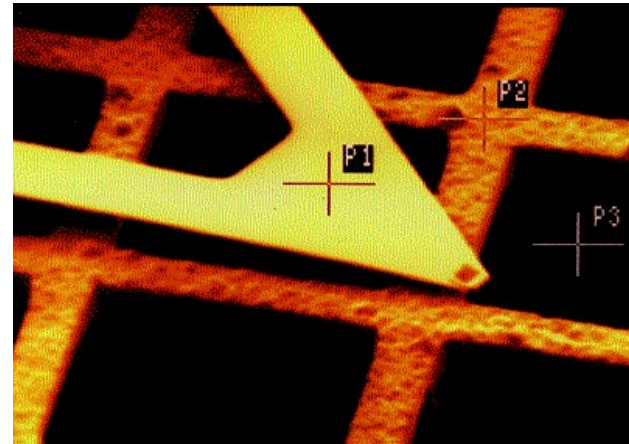
FIG. 10. A negative ion static SIMS spectrum of poly(2-hydroxyethyl methacrylate) with the primary peaks identified.

TABLE 5 Analytical Capabilities of SIMS

	Static SIMS	Dynamic SIMS
Identify hydrogen	✓	✓
Identify other elements (often must be inferred from the data)	✓	✓
Suggest molecular structures (inferred from the data)	✓	
Observe extremely high mass fragments (proteins, polymers)	✓	
Detection of extremely low concentrations	✓	✓
Depth profile to 1 μm into the sample		✓
Observe the outermost 1–2 atomic layers	✓	
High spatial resolution (features as small as approximately 500 Å)	✓	✓
Semiquantitative analysis (for limited sets of specimens)	✓	
Useful for polymers	✓	
Useful for inorganics (metals, ceramics, etc.)	✓	✓
Useful for powders, films, fibers, etc.	✓	✓

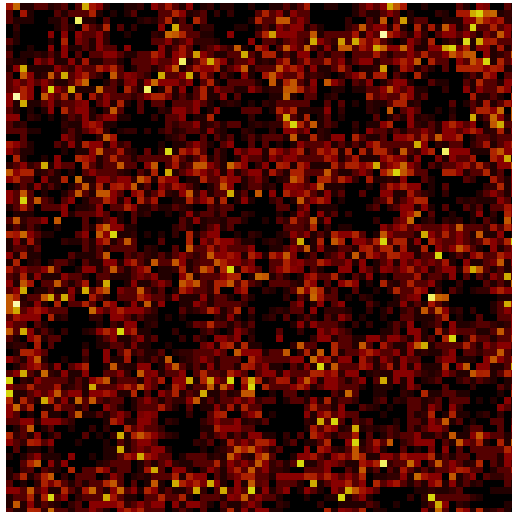
Imaging Modes for ESCA and SIMS

Auger image of AFM cantilever over inorganic pattern – point in the image can be selected for greater analysis

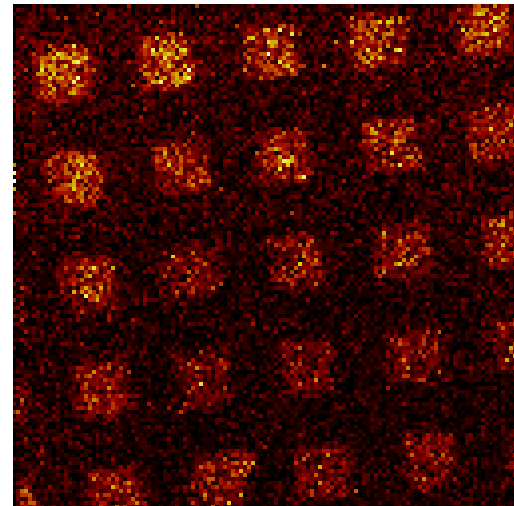


ToF-SIMS Image: Fibronectin Squares on PEO background

CH_3O^+ ion image



CN^- ion image

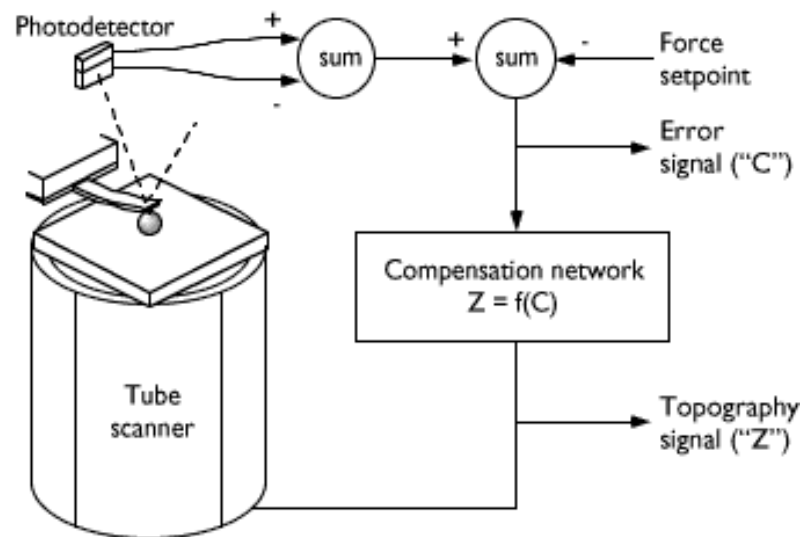
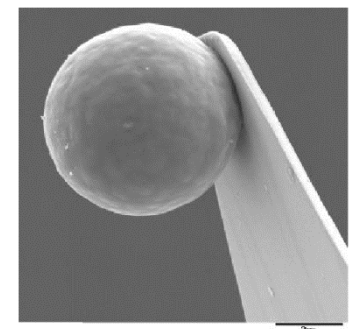
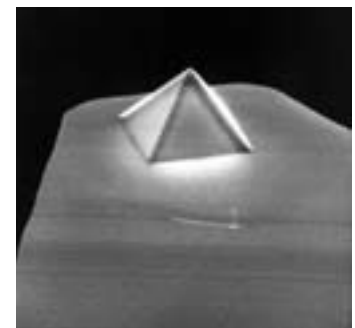
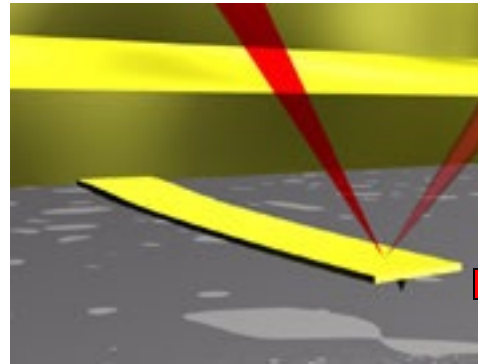
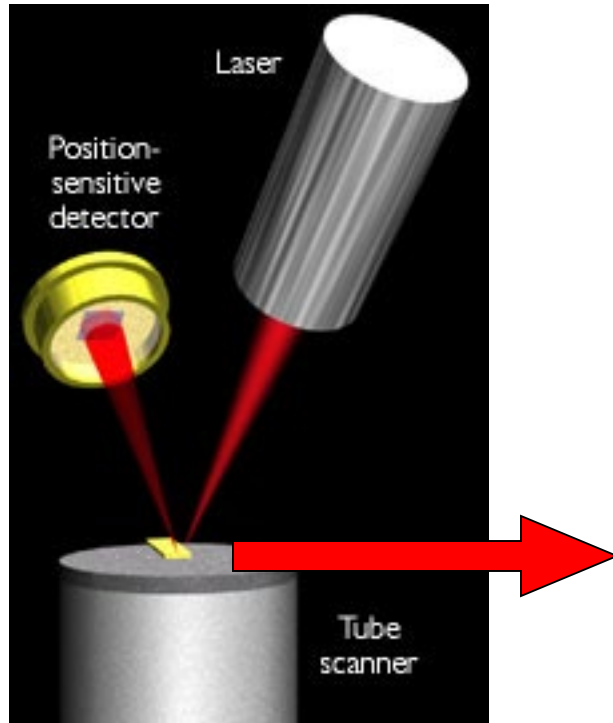


SPM

Alias:	Scanning Probe Microscopy, Scanning Force M.
Modes:	Lots: atomic force (AFM) , tunneling (STM), magnetic, kelvin probe, electrostatic, acoustical, calorimetry
Probe:	Cantilever tip – single atom!
Signal:	Position of tip, etc
Information:	Topography, etc
Sample:	Just about anything
Principle:	Raster a small tip over the surface to collect data and reconstruct image
Depth:	5 or less Å
Spatial Resolution:	as low as 1 Å ²
Sensitivity:	Atomic sensitivity
Relative Cost:	Moderate to expensive
Other:	Nearfield Scanning Optical Microscopy (NSOM) Tip functionalization (change force regime or promote specific binding)

Instrument

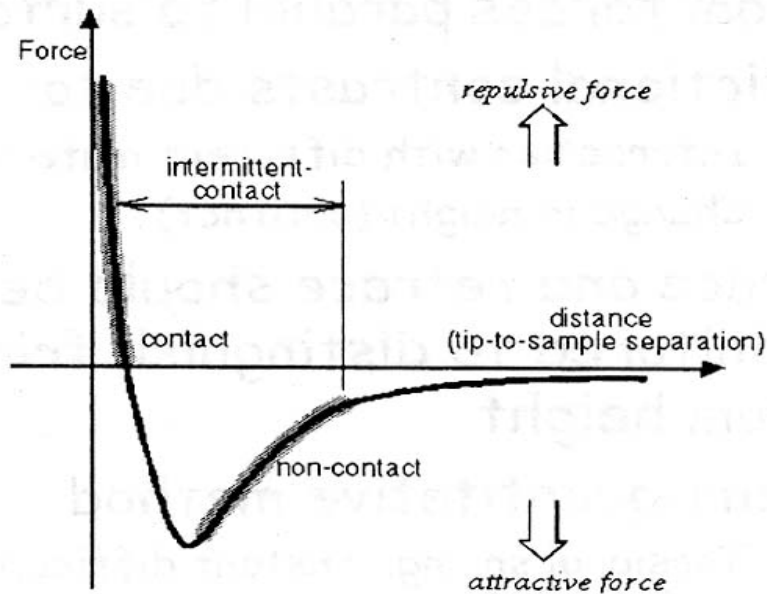
As the cantilever moves the position of the reflected laser beam on the detector changes



It is necessary to "close" the control loop to maintain the signal

$$10^{-8} - 10^{-6} \text{ N}$$

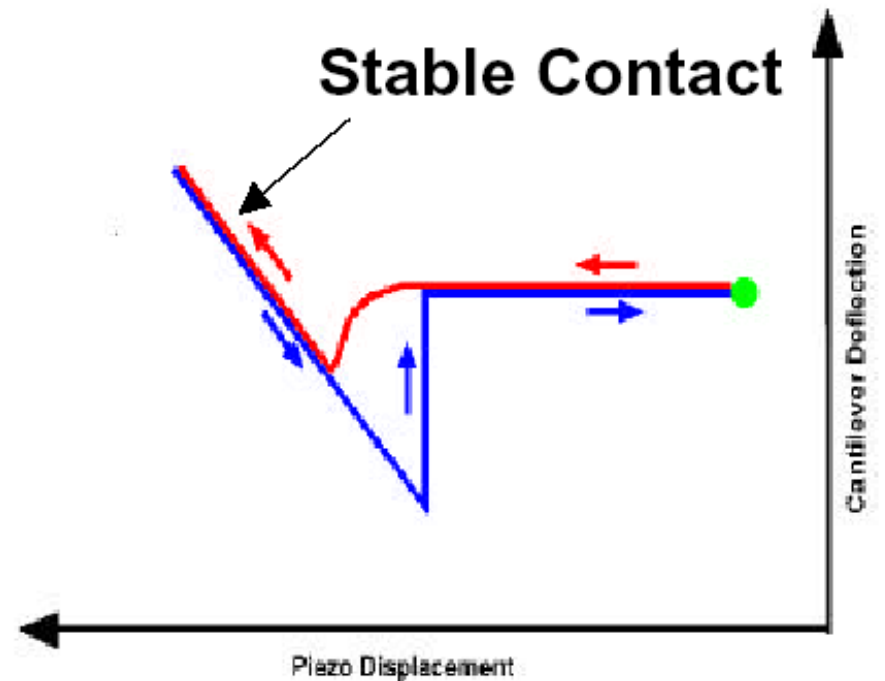
AFM



Thus one can stably locate the tip “in contact” with the surface

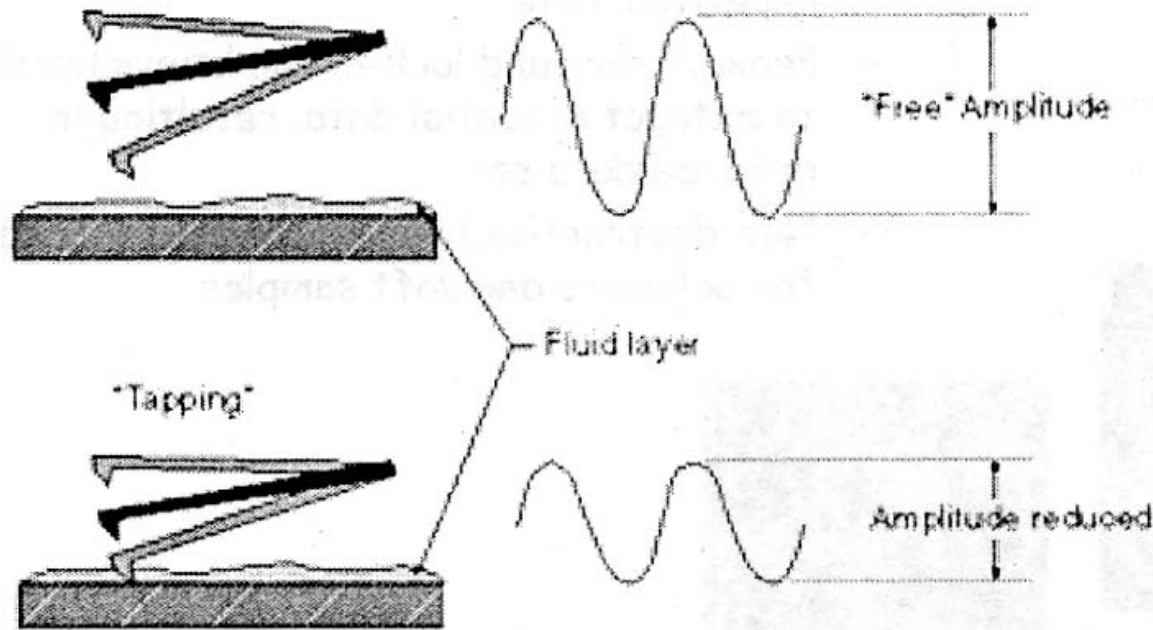
Main AFM modes: **Contact**, **Non-Contact**, **Tapping**, Adhesion Force, Lateral Force

DLVO theory (among others) is used to quantify “force curve” data



→ approach → retraction

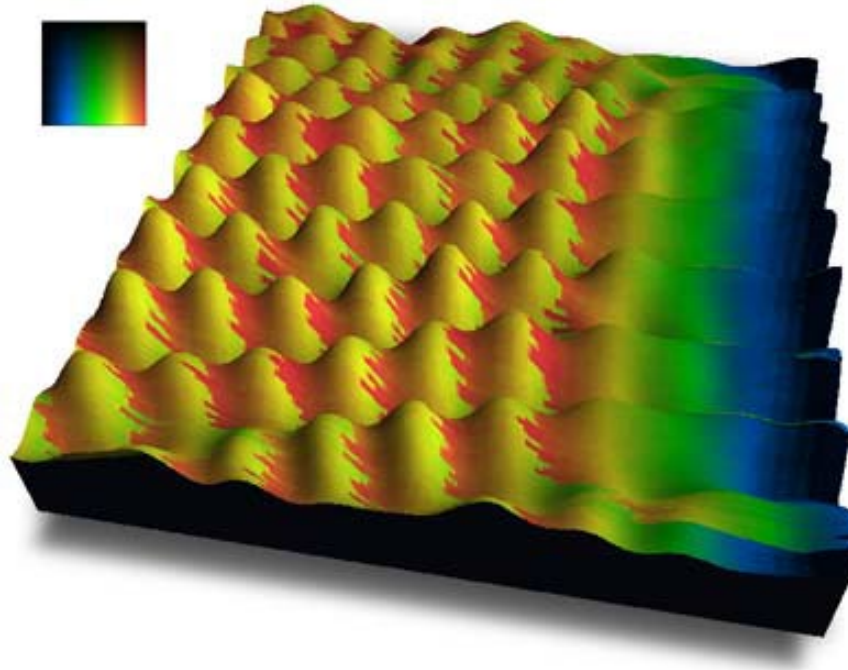
AFM Tapping Mode



If a particular sample seems to be having problems with adhesive capillary tip-sample forces, these **forces can be substantially reduced by immersing the tip and the sample in a liquid**, which generally requires the use of a liquid cell accessory. Further reduction of the attractive forces is possible by operating **in appropriate solvents, especially ethanol**. The liquid cell is also used for in-situ electrochemical and biological experiments.

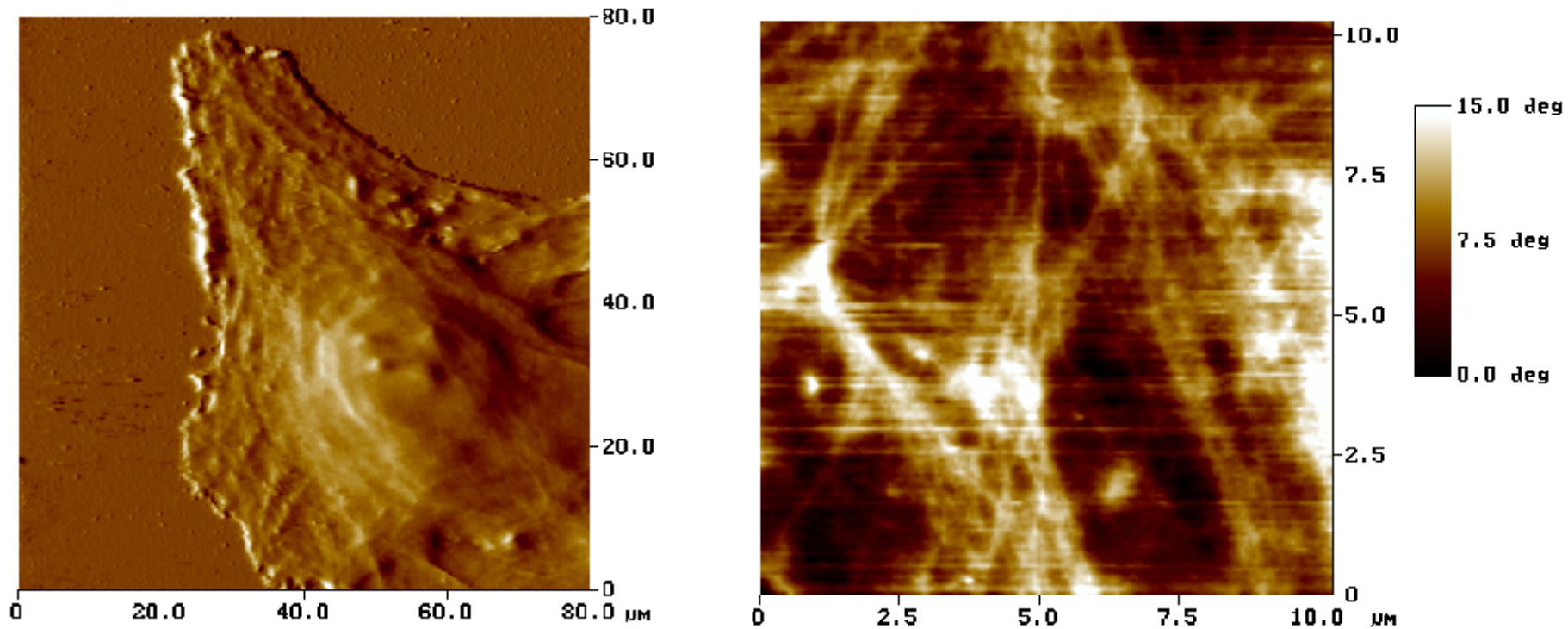
Conventional liquid cells have a drive frequency of about 10 kHz. This matches the resonance frequency of cantilevers that vibrate in air at about 50 - 70 kHz. Typical values of the Q-factor in liquids are 1 - 3 due to increased hydrodynamic damping. **Rectangular cantilevers suffer less damping in liquids than triangular cantilevers.**

Nice, Pretty Pictures



2.5 x 2.5 nm simultaneous topographic and friction image of highly oriented pyrolytic graphic (HOPG). The bumps represent the topographic atomic corrugation, while the coloring reflects the lateral forces on the tip. The scan direction was right to left.

Nice, Pretty Pictures



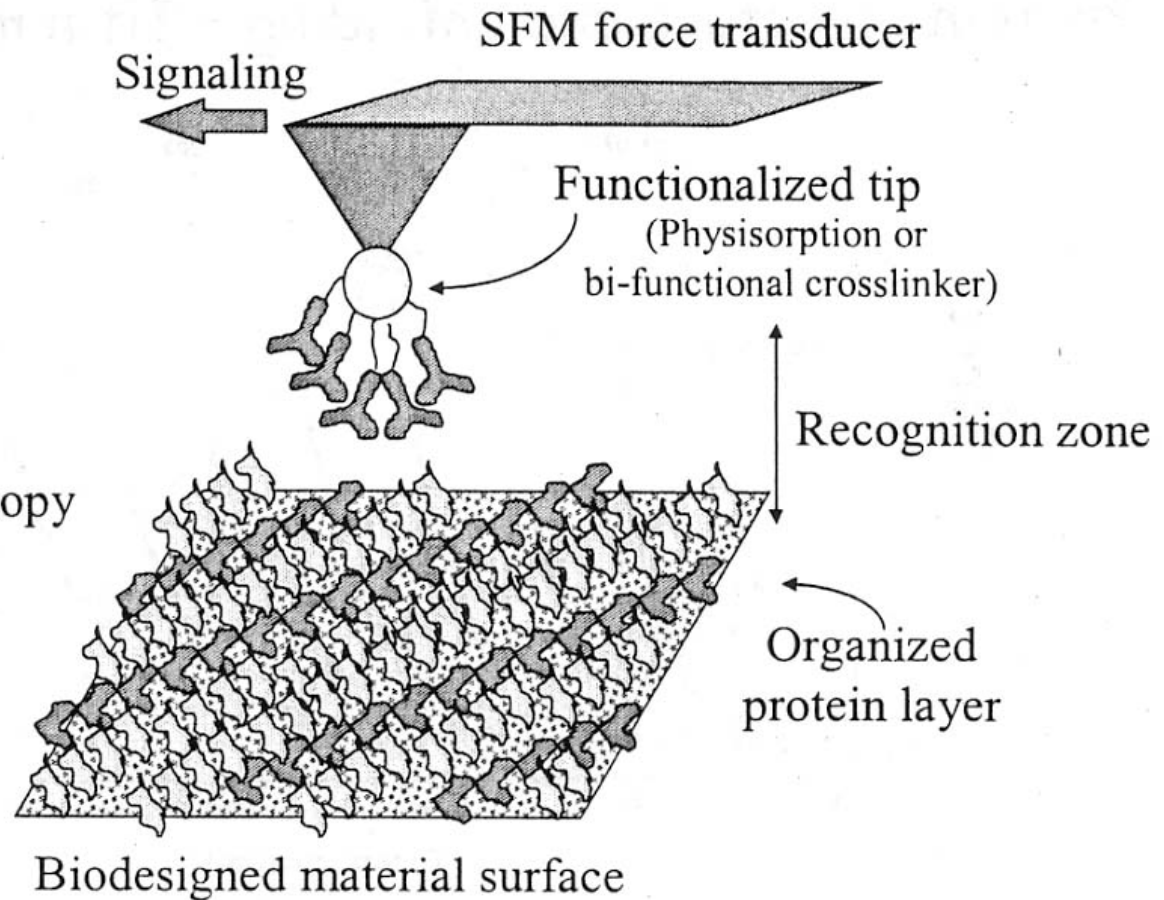
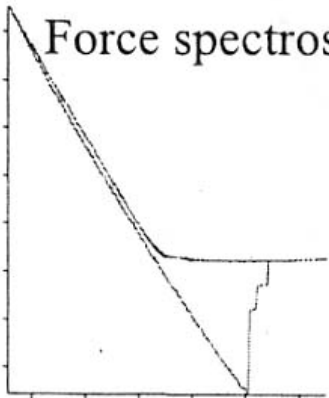
AFM topography and phase mode images of live cells

SFM: Functional Imaging

Specific imaging



Force spectroscopy



AFM – Caution!

There is a great tendency to “see what you want” to in AFM images, although multimode operation helps to reduce interpretations. This technique has not-so-obvious limitations:

- Tip contamination
- Piezo non-linearities and drift
- Tips are rarely characterized for spring constant, geometry
- Artifacts (double tip)
- topography / phase convolution in lateral force
- “near sightedness”

Etc: Optical Techniques

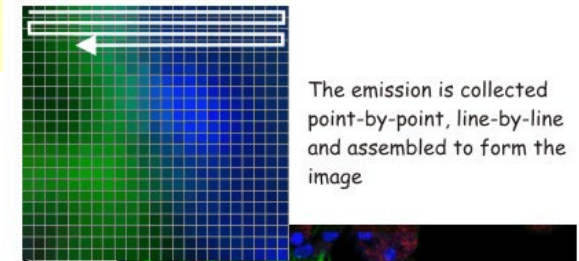
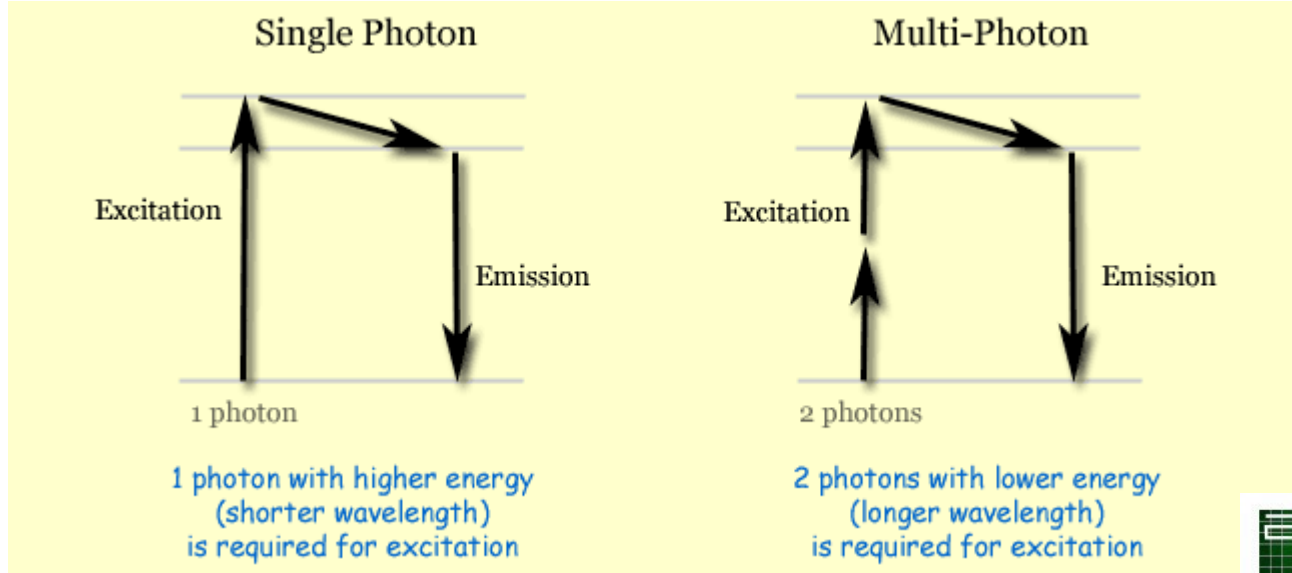
All of the usual complement of microscopic modalities, plus:

Sum Frequency Generation (SFG): Use non-linear excitation at any asymmetry (i.e. interface) to gather IR vibrational and Raman data. Has sub-monolayer sensitivity and gives orientation information.

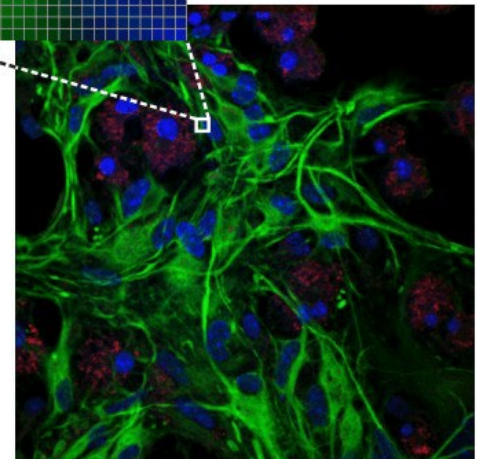
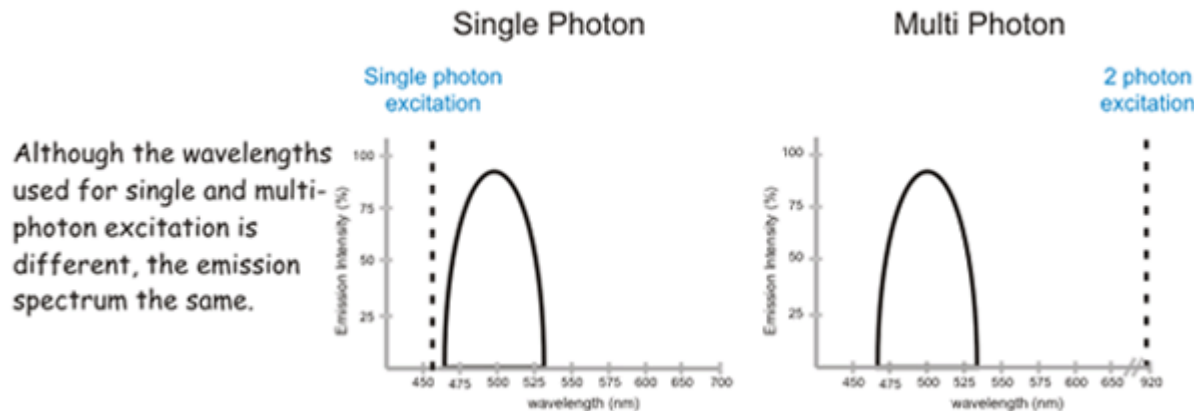
Multiphoton Techniques: Increased depth resolution and greater depth penetration into tissue with increased signal to noise. Not inherently depth specific – although it can be applied to membranes and diffuse interfaces (e.g. fluorescence correlation spectroscopy).

Evanescent wave techniques: Creates surface-specific excitation (e.g. TIRF) or collects information from near surface (e.g. SPR) by creating an evanescent wave. Very useful for binding or adsorption studies with biological systems.

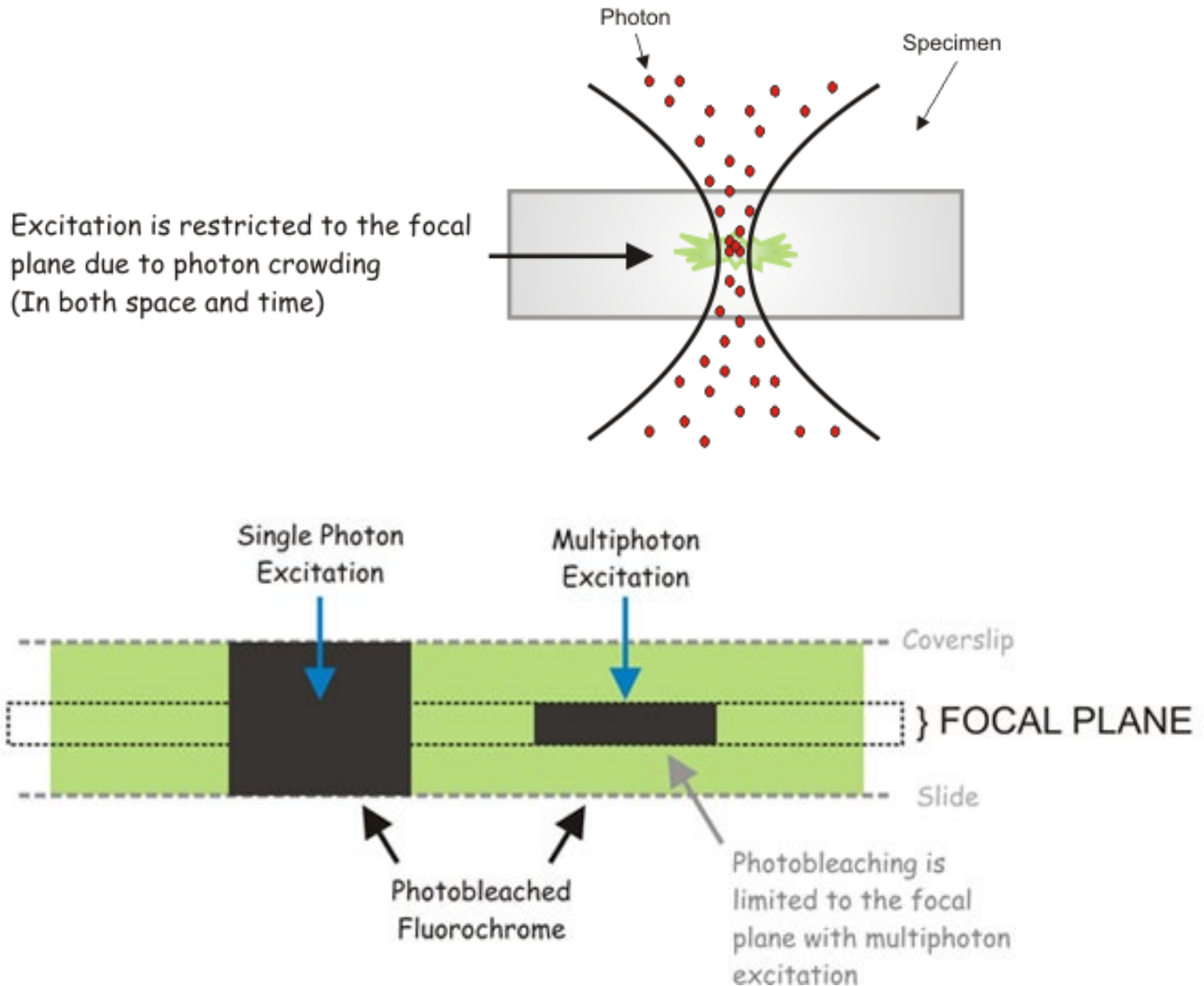
Brief introduction of multiphoton



The emission is collected point-by-point, line-by-line and assembled to form the image



Brief introduction of multiphoton



[http://www.bioen.utah.edu/faculty/pat/Courses/biomaterials/
coursenotes.html](http://www.bioen.utah.edu/faculty/pat/Courses/biomaterials/coursenotes.html)